

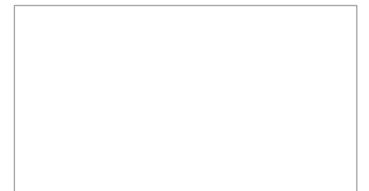
# QpiAI™ - Joint AI Training and Certification Program

Bayesian Inference

Sourjya Sarkar

# Bayesian Inference

- **Bayesian Modelling**
- Priors
- MLE and MAP
- Bayesian Inference Examples
- Tractability and Approximation Methods
- Variational Inference
- Evidence Lower Bound (ELBO)
- Kullback-Liebler (KL) Divergence
- Mean field Variational Inference
- Variational Bayes Learning
- Myths about Bayesian Methods



# Bayesian Modelling

Everything follows from two simple rules:

**Sum rule:**  $P(x) = \sum_y P(x, y)$

**Product rule:**  $P(x, y) = P(x)P(y|x)$

$$P(\theta|\mathcal{D}, m) = \frac{P(\mathcal{D}|\theta, m)P(\theta|m)}{P(\mathcal{D}|m)}$$

$$P(\mathcal{D}|m) = \int P(\mathcal{D}|\theta, m)P(\theta|m)d\theta$$

$P(\mathcal{D}|\theta, m)$  likelihood of parameters  $\theta$  with data  $\mathcal{D}$  in model  $m$

$P(\theta|m)$  prior probability of  $\theta$

$P(\theta|\mathcal{D}, m)$  posterior probability of  $\theta$  given data  $\mathcal{D}$

**Prediction (Posterior predictive):**

$$P(x|\mathcal{D}, m) = \int P(x|\theta, \mathcal{D}, m)P(\theta|\mathcal{D}, m)d\theta$$

**Model Comparison:**

$$P(m|\mathcal{D}) = \frac{P(\mathcal{D}|m)P(m)}{P(\mathcal{D})}$$

$$P(\mathcal{D}|m) = \int P(\mathcal{D}|\theta, m)P(\theta|m)d\theta$$

# A Simple Example: Learning from Gaussians

## Clustering with Gaussian Mixtures (Density Estimation)

**Data:**  $\mathcal{D} = \{x^{(n)}\}$  for  $n=1, \dots, N$

$$x^{(n)} \in \mathbb{R}^D$$

**Model:**

$$x^{(n)} \sim \sum_{i=1}^m \pi_i p_i(x^{(n)})$$

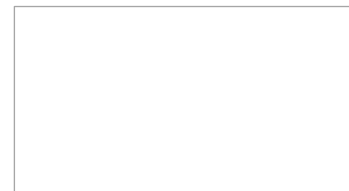
Where

$$p_i(x^{(n)}) = \mathcal{N}(\mu^{(i)}, \Sigma^{(i)})$$

**Parameters:**

$$\theta = ((\mu^{(1)}, \Sigma^{(1)}) \dots, (\mu^{(m)}, \Sigma^{(m)}), \pi)$$

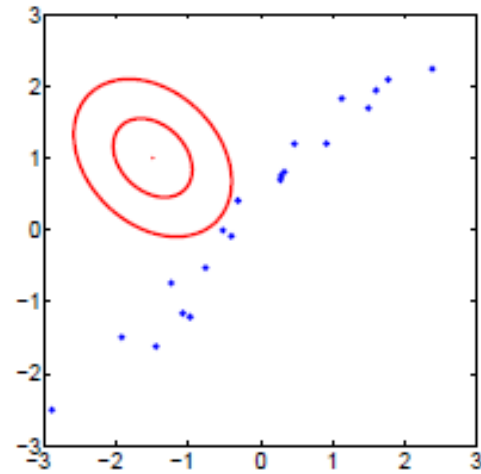
**Goal:** To infer  $\theta$  from the data, predict the density  $p(x|\mathcal{D}, m)$ , and infer which points belong to the same cluster



# A Simple Example: Learning a Gaussian

$$P(\theta|\mathcal{D}, m) = \frac{P(\mathcal{D}|\theta, m)P(\theta|m)}{P(\mathcal{D}|m)}$$

- The model  $m$  is a multivariate Gaussian.
- Data,  $\mathcal{D}$  are the blue dots.
- Parameters  $\theta$  are the mean vector and covariance matrix of the Gaussian.



# Bayesian Occam's Razor and Model Selection

Compare model classes, e.g.  $m$  and  $m'$ , using posterior probabilities given  $\mathcal{D}$ :

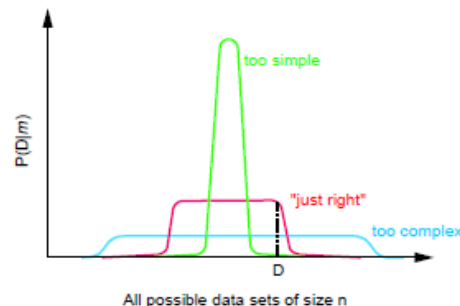
$$P(m|\mathcal{D}) = \frac{P(\mathcal{D}|m)P(m)}{P(\mathcal{D})}, \quad p(\mathcal{D}|m) = \int p(\mathcal{D}|\theta, m)p(\theta|m)d\theta$$

## Interpretations of the Marginal Likelihood (“model evidence”):

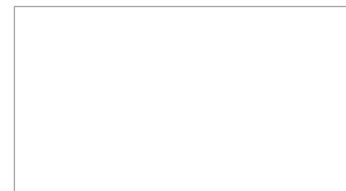
- The probability that randomly selected parameters from the prior would generate  $\mathcal{D}$ .
- Probability of the data under the model, averaging over all possible parameter values.

Model classes that are **too simple** are unlikely to generate the data set.

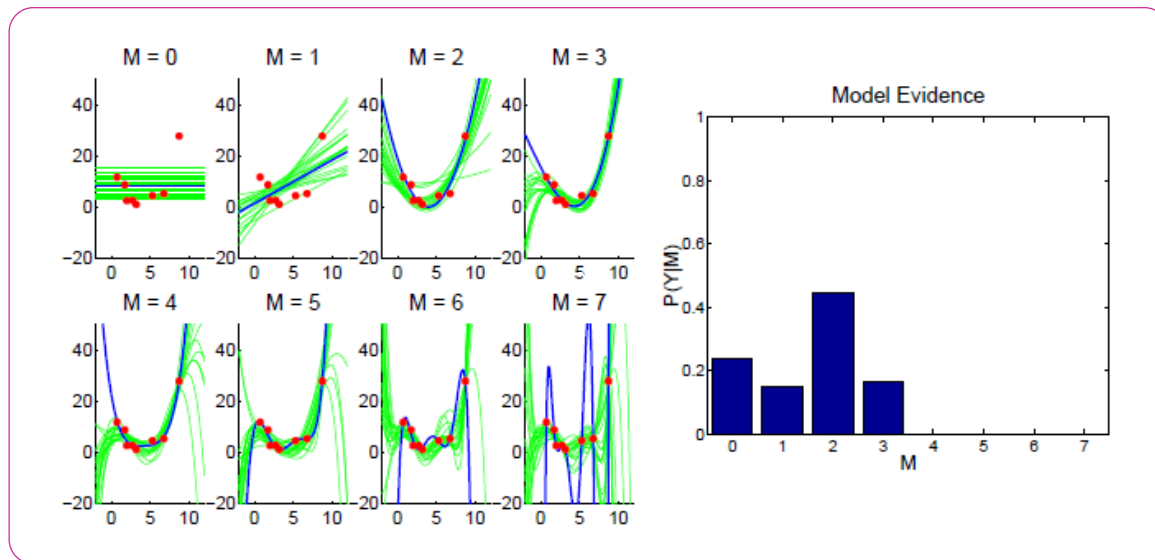
Model classes that are **too complex** can generate many possible data sets, so again, they are unlikely to generate that particular data set at random.



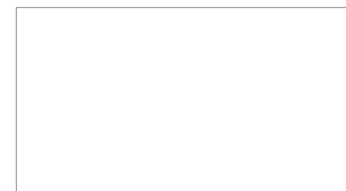
“Accept the simplest explanation that fits the data”



# Bayesian Model Selection: Occam's Razor at Work



For example, for quadratic polynomials ( $m = 2$ ):  $y = a_0 + a_1x + a_2x^2 + \epsilon$ , where  $\epsilon \sim \mathcal{N}(0, \sigma^2)$  and parameters  $\theta = (a_0 a_1 a_2 \sigma)$



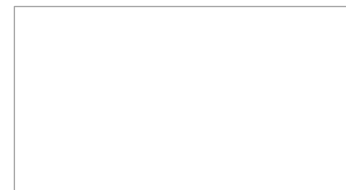
# Taxonomy of ML Models

## Generative Methods

- Model class-conditional pdfs and prior probabilities
- “Generative” since sampling can generate synthetic data points
- Popular models
  - Gaussians, Naïve Bayes, Mixture of multinomials
  - Mixture of Gaussians, Mixtures of experts, Hidden Markov Models (HMM)
  - Sigmoidal belief networks, Bayesian networks Markov random fields

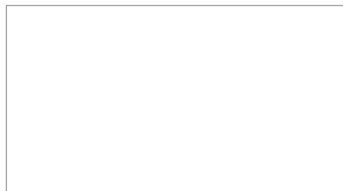
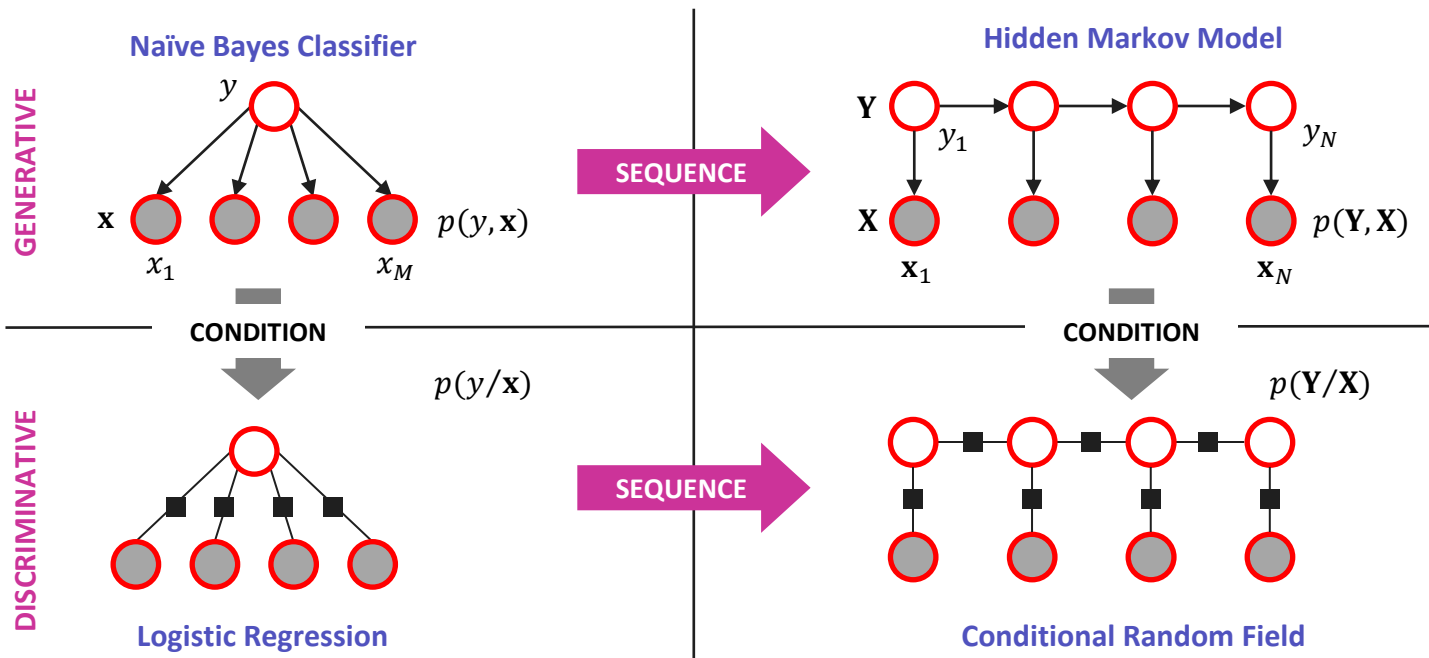
## Discriminative Methods

- Directly estimate posterior probabilities
- No attempt to model underlying probability distributions
- Focus computational resources on given task for better performance
- Popular models
  - Logistic regression, SVMs
  - Traditional neural networks
  - Conditional Random Fields (CRF)
  - Gaussian Processes



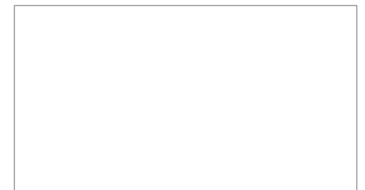


# Graphical Model Relationship



# Bayesian Inference

- Bayesian Modelling
- **Priors**
- MLE and MAP
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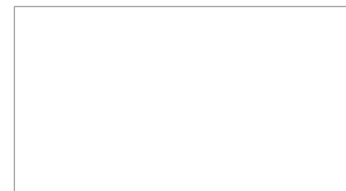


# On Choosing Priors

- **Objective Priors:** noninformative priors that attempt to capture ignorance and have good frequentist properties.
- **Subjective Priors:** priors should capture our beliefs as well as possible. They are subjective but not arbitrary.
- **Hierarchical Priors:** multiple levels of priors:

$$\begin{aligned} P(\theta) &= \int p(\theta|\alpha)p(\alpha)d\alpha \\ &= \int p(\theta|\alpha) \int p(\alpha|\beta)p(\beta)d\beta d\alpha \end{aligned}$$

- **Empirical Priors:** learn some of the parameters of the prior from the data (“Empirical Bayes”)



# Objective Priors

## Non-informative Priors:

Consider a Gaussian with mean  $\mu$  and variance  $\sigma^2$ .

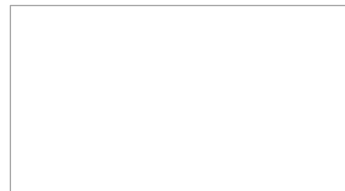
The parameter  $\mu$  informs about the location of the data.

If we pick  $p(\mu) = p(\mu - a) \forall a$  then predictions are location invariant

$$p(x|x') = p(x - a|x' - a)$$

- But  $p(\mu) = p(\mu - a) \forall a$  implies  $p(\mu) = \text{Unif}(-\infty, \infty)$  which is **improper**.
- Similarly,  $\sigma$  informs about the scale of the data, so we can pick  $p(\sigma) \propto 1/\sigma$

**Problems:** It is hard (impossible) to generalize to all parameters of a complicated model. Risk of incoherent inferences (e.g.  $E_x E_y [Y|X] \neq E_y [Y]$ ), paradoxes, and improper posteriors.



# Objective Priors

## Reference Priors:

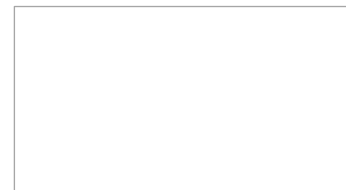
Captures the following notion of non informativeness. Given a model  $p(x|\theta)$  we wish to find the prior on  $\theta$  such that an experiment involving observing  $x$  is expected to provide the most information about  $\theta$ .

That is, most of the information about  $\theta$  will come from the experiment rather than the prior. The information about  $\theta$  is:

$$I(\theta|x) = - \int p(\theta) \log p(\theta) d\theta - \left( - \int p(\theta, x) \log p(\theta|x) d\theta dx \right)$$

This can be generalized to experiments with  $n$  observations (giving different answers)

**Problems:** Hard to compute in general (e.g. MCMC schemes), prior depends on the size of data to be observed.



# Objective Priors

## Jeffreys Priors:

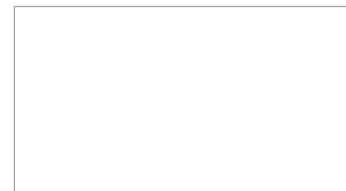
Motivated by invariance arguments: the principle for choosing priors should not depend on the parameterization.

$$p(\phi) = p(\theta) \left| \frac{d\theta}{d\phi} \right|$$

$$p(\theta) \propto h(\theta)^{1/2}$$

$$h(\theta) = - \int p(x|\theta) \frac{\partial^2}{\partial \theta^2} \log p(x|\theta) dx \quad (\text{Fisher information})$$

**Problems:** It is hard (impossible) to generalize to all parameters of a complicated model. Risk of incoherent inferences (e.g.  $E_x E_y[Y|X] \neq E_y[Y]$ ), paradoxes, and improper posteriors.



# Subjective Priors

Priors should capture our beliefs as well as possible.

Otherwise we are not coherent.

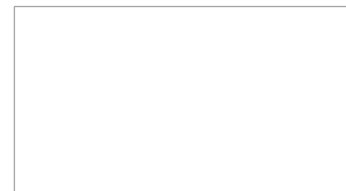
How do we know our beliefs?

- Think about the problem domain.
- Generate data from the prior. Does it match expectations?

Even very vague priors beliefs can be useful, since the data will concentrate the posterior around reasonable models.

The key ingredient of Bayesian methods is not the prior, it's the idea of averaging over different possibilities.

**Problems:** May be swamped by data if the likelihood does not conform to the pre-specified prior.  
Requires experience and domain knowledge



# Empirical Priors

Consider a hierarchical model with parameters  $\theta$  and hyperparameters  $\alpha$

$$p(\mathcal{D}|\alpha) = \int p(\mathcal{D}|\theta)p(\theta|\alpha)d\theta$$

Estimate hyperparameters from the data

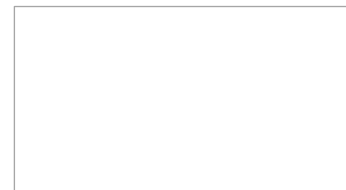
$$\hat{\alpha} = \underset{\alpha}{\operatorname{argmax}} p(\mathcal{D}|\alpha) \text{ (MLE-II)}$$

**Prediction:**

$$p(x|\mathcal{D}, \hat{\alpha}) = \int p(x|\theta)p(\theta|\mathcal{D}, \hat{\alpha})d\theta$$

**Advantages:** Robust – overcomes some limitations of mis-specification of the prior.

**Problem:** Double counting of data / overfitting.





# Exponential Family and Conjugate Priors

$p(x|\theta)$  in the **Exponential family** if it can be written as:

$$p(x|\theta) = f(x)g(\theta) \exp\{\phi(\theta)^T s(x)\}$$

$\theta$  vector of natural parameters

$s(x)$  vector of sufficient statistics

$f$  and  $g$  positive function of  $x$  and  $\theta$ , respectively.

The **conjugate prior** for this is

$$p(\theta) = h(\eta, \nu) g(\theta)^\eta \exp\{\phi(\theta)^T \nu\}$$

Where  $\eta$  and  $\nu$  are hyperparameters and  $h$  is the normalizing function.

The posterior for  $N$  data points is also conjugate (by definition), with hyperparameters  $\eta + N$  and  $\nu + \sum_n s(x_n)$ .

This is **computationally convenient**.

$$p(\theta|x_1, \dots, x_N) = h\left(\eta + N, \nu + \sum_n s(x_n)\right) g(\theta)^{\eta+N} \exp\left\{\phi(\theta)^T \left(\nu + \sum_n s(x_n)\right)\right\}$$

## A prior exercise: logistic regression

Consider logistic regression: 1

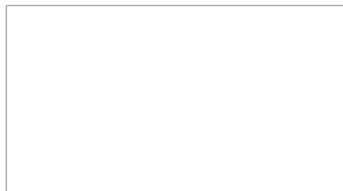
$$p(y = 1|w, b, x) = p = \frac{1}{1 + \exp\{-w^T x + b\}}$$

Assume  $x \in \{0,1\}^D$ . How should we choose the prior  $p(w, b)$ ?

For simplicity, let's focus on the case:

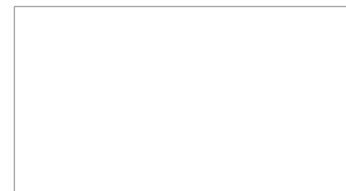
$$p(w) = N(0, \sigma^2 I) \quad p(b) = N(0, \sigma^2)$$

- Q:** For  $D = 1$ , what happens if we choose  $\sigma \gg 0$  as a “non-informative” prior?
- Q:** How does  $p$  behave for very small  $\sigma$ ?
- Q:** Repeat for  $D = 10$ .
- Q:** Repeat for  $D = 1000$  where we expect  $x$  to be sparse (with density 1%).
- Q:** Can you think of a hierarchical prior which captures a range of desired behaviours?



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# MLE vs. MAP

Maximum Likelihood estimation (MLE)

Choose value that maximizes the probability of observed data

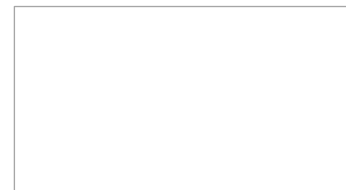
$$\hat{\theta}_{MLE} = \arg \max_{\theta} P(D|\theta)$$

Maximum a posteriori (MAP) estimation

Choose value that is most probable given observed data and prior belief

$$\begin{aligned}\hat{\theta}_{MAP} &= \arg \max_{\theta} P(\theta|D) \\ &= \arg \max_{\theta} P(D|\theta) P(\theta)\end{aligned}$$

When is MAP same as MLE?



# MLE vs. MAP

Suppose we have data  $\mathcal{D} = \{x^{(i)}\}_{i=1}^N$

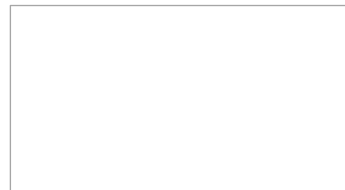
$$\theta^{MLE} = \arg \max_{\theta} \prod_{i=1}^N p(x^{(i)} | \theta)$$

Maximum Likelihood  
Estimate (MLE)

$$\theta^{MAP} = \arg \max_{\theta} \prod_{i=1}^N p(x^{(i)} | \theta) \underbrace{p(\theta)}$$

Maximum a posteriori  
(MAP) estimate

**Prior**



# MAP estimation for Binomial Distribution

**Coin flip problem:** Likelihood is Binomial

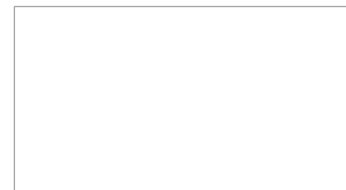
$$P(\mathcal{D}|\theta) = \binom{n}{\alpha_H} \theta^{\alpha_H} (1 - \theta)^{\alpha_T}$$

If the prior is Beta distribution,

$$P(\theta) = \frac{\theta^{\beta_H-1} (1 - \theta)^{\beta_T-1}}{B(\beta_H, \beta_T)} \sim \text{Beta}(\beta_H, \beta_T)$$

⇒ Posterior is Beta distribution

$$\text{Beta Function: } B(x, y) = \int_0^1 t^{x-1} (1 - t)^{y-1} dt$$



# MAP estimation for Binomial Distribution

Likelihood is Binomial

$$P(\mathcal{D}|\theta) = \binom{n}{\alpha_H} \theta^{\alpha_H} (1 - \theta)^{\alpha_T}$$

Prior is Beta distribution:

$$P(\theta) = \frac{\theta^{\beta_H-1} (1 - \theta)^{\beta_T-1}}{B(\beta_H, \beta_T)} \sim \text{Beta}(\beta_H, \beta_T)$$

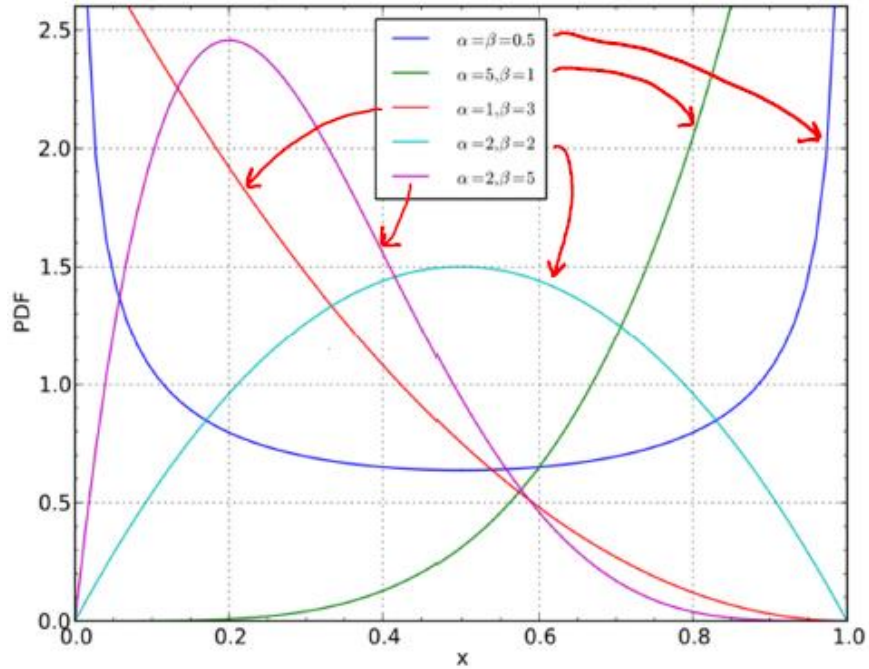
⇒ Posterior is Beta distribution

$$P(\theta|D) \sim \text{Beta}(\beta_H + \alpha_H, \beta_T + \alpha_T)$$

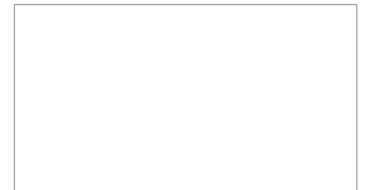
$P(\theta)$  and  $P(\theta|D)$  have the same form! [Conjugate prior]

$$\begin{aligned} \hat{\theta}_{MAP} &= \arg \max_{\theta} P(\theta|D) = \arg \max_{\theta} P(D|\theta) P(\theta) \\ &= \frac{\alpha_H + \beta_H - 1}{\alpha_H + \beta_H + \alpha_T + \beta_T - 2} \end{aligned}$$

# Beta Distribution



More concentrated as values of  $\alpha, \beta$  increase

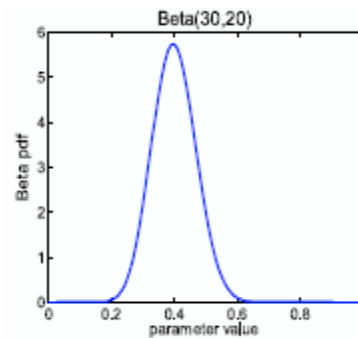
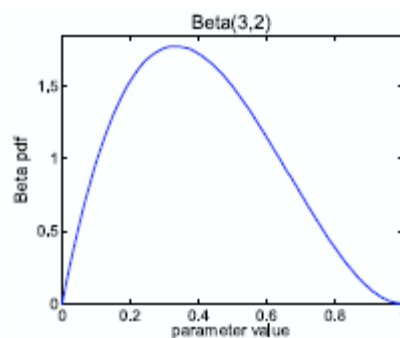
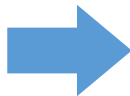
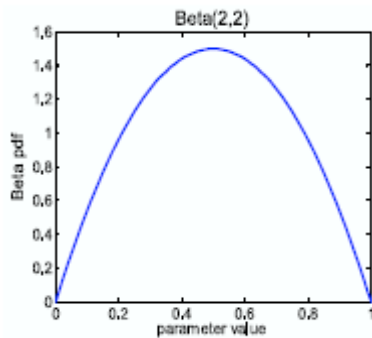




# Beta Conjugate Prior

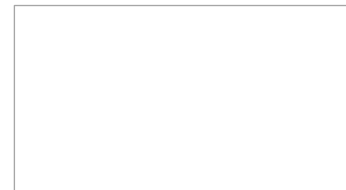
$$P(\theta) \sim \text{Beta}(\beta_H, \beta_T)$$

$$P(\theta|D) \sim \text{Beta}(\beta_H + \alpha_H, \beta_T + \alpha_T)$$



As  $n = \alpha_H + \alpha_T$   
increases

As we get more samples, effect of prior is “washed out”



# From Binomial to Multinomial

**Example:** Dice roll problem (6 outcomes instead of 2)

Likelihood is  $\sim$  Multinomial ( $\theta = \{\theta_1, \theta_2, \dots, \theta_k\}$ )

$$P(\mathcal{D}|\theta) = \theta_1^{\alpha_1} \theta_2^{\alpha_2} \dots \theta_k^{\alpha_k}$$

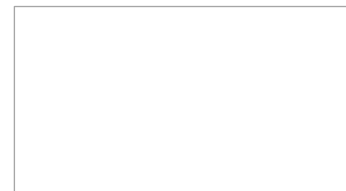
If prior is Dirichlet distribution,

$$P(\theta) = \frac{\prod_{i=1}^{\kappa} \theta_i^{\beta_i}}{B(\beta_1, \dots, \beta_{\kappa})} \sim \text{Dirichlet}(\beta_1, \dots, \beta_{\kappa})$$

Then posterior is Dirichlet distribution,

$$P(\theta|\mathcal{D}) \sim \text{Dirichlet}(\beta_1 + \alpha_1, \dots, \beta_{\kappa} + \alpha_{\kappa})$$

**For Multinomial, conjugate prior is Dirichlet distribution.**



# Maximum Likelihood Estimate (MLE)

## Estimating a parameter

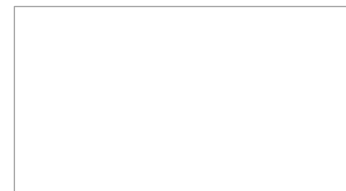
**Example:** Suppose we want to know the percentage  $p$  of people for whom cilantro tastes like soap.

**Experiment:** Ask  $n$  random people to taste cilantro.

**Model:**  $X_i \sim \text{Bernoulli}(p)$  is whether the  $i^{\text{th}}$  person says it tastes like soap.

**Data:**  $x_1, \dots, x_n$  are the results of the experiment

**Inference:** Estimate  $p$  from the data.

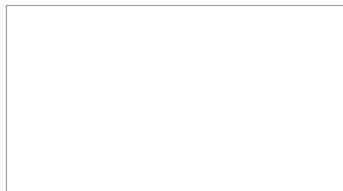


# Maximum Likelihood Estimate (MLE)

## Parameter of interest

**Example:** You ask 100 people to taste cilantro and 55 say it tastes like soap. Use this data to estimate  $p$  the fraction of all people for whom it tastes like soap.

So,  $p$  is the parameter of interest.



# Maximum Likelihood Estimate (MLE)

## Likelihood

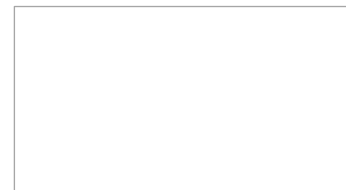
For a given value of  $p$  the probability of getting 55 'successes' is the binomial probability

$$P(55 \text{ soap}|p) = \binom{100}{55} p^{55} (1 - p)^{45}$$

### Definition:

$$\text{The likelihood } P(\text{data}|p) = \binom{100}{55} p^{55} (1 - p)^{45}$$

**NOTICE:** The likelihood takes the data as fixed and computes the probability of the data for a given  $p$ .



# Maximum Likelihood Estimate (MLE)

## Maximum likelihood estimate (MLE)

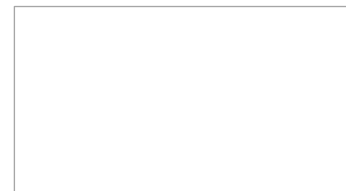
The maximum likelihood estimate (MLE) is a way to estimate the value of a **parameter of interest**

The MLE is the value of  $p$  that **maximizes** the likelihood

Different problems call for **different methods** of finding the maximum

Here are two – there are others

1. Calculus: To find the MLE, solve  $\frac{d}{dp}P(data|p) = 0$  for  $p$ . (We should also check that the critical point is a maximum)
2. Sometimes the derivative is never 0 and the MLE is at an endpoint of the allowable range.



# Maximum Likelihood Estimate (MLE)

## Cilantro tasting MLE

The MLE for the cilantro tasting experiment is found by calculus

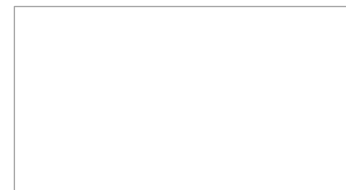
$$\frac{dP(\text{data}|p)}{dp} = \binom{100}{55} (55p^{54}(1-p)^{45} - 45p^{55}(1-p)^{44}) = 0$$

A sequence of algebraic steps gives:

$$\begin{aligned} 55p^{54}(1-p)^{45} &= 45p^{55}(1-p)^{44} \\ 55(1-p) &= 45p \\ 55 &= 100p \end{aligned}$$

Therefore the MLE is

$$\hat{p} = \frac{55}{100}$$



# Maximum Likelihood Estimate (MLE)

## Log likelihood

Because the log function turns multiplication into addition it is often convenient to use the log of the likelihood function

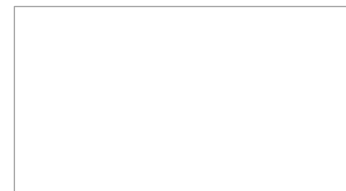
Example

$$\log \text{likelihood} = \ln(\text{likelihood}) = \ln(P(\text{data}|P))$$

$$\text{Likelihood } P(\text{data}|p) = \binom{100}{55} p^{55} (1-p)^{45}$$

$$\text{Log likelihood} = \ln \left( \binom{100}{55} \right) + 55 \ln(p) + 45 \ln(1-p)$$

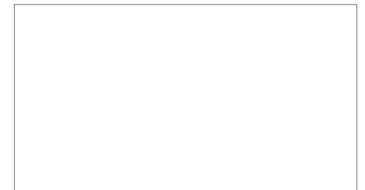
Note first term is just a constant





# Bayesian Inference

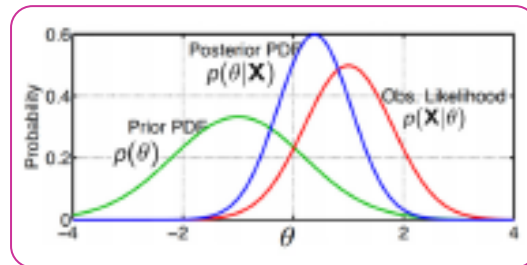
- Bayesian Modelling
- Priors
- MLE and MAP
- **Bayesian Inference Examples**
- Tractability and Approximation Methods
- Variational Inference
- Evidence Lower Bound (ELBO)
- Kullback-Liebler (KL) Divergence
- Mean field Variational Inference
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- Myths about Bayesian Methods



# Recap: Bayesian Inference

- Given data  $\mathbf{X}$  from a model  $m$  with parameters  $\theta$ , the posterior over the parameters  $\theta$

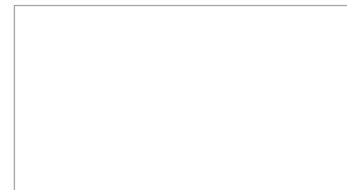
$$p(\theta|\mathbf{X}, m) = \frac{p(\mathbf{X}, \theta|m)}{p(\mathbf{X}|m)} = \frac{p(\mathbf{X}|\theta, m)p(\theta|m)}{\int p(\mathbf{X}|\theta, m)p(\theta|m)d\theta} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Marginal likelihood}}$$



- Can use the posterior for various purposes, e.g.,
  - Getting point estimates e.g., mode (though, for this, directly doing point estimation is often easier)
  - Uncertainty in our estimates of  $\theta$  (variance, credible intervals, etc.)
  - Computing the **posterior predictive distribution** (PPD) for new data, e.g.,

$$p(x_*|\mathbf{X}, m) = \int p(x_*|\theta, m)p(\theta|\mathbf{X}, m)d\theta$$

- Caveat: Computing the posterior/PPD is in general hard (due to the intractable integrals involved)

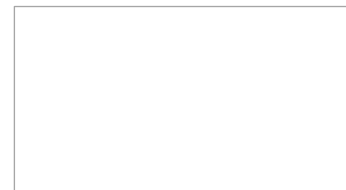


# Marginal Likelihood and Its Usefulness

- Likelihood vs Marginal Likelihood:  $p(\mathbf{X}|\theta, m)$  vs  $p(\mathbf{X}|m)$ 
  - Prob. of  $\mathbf{X}$  for a single  $\theta$  under model  $m$  vs prob. Of  $\mathbf{X}$  averaged over all  $\theta$ 's under model  $m$
- Can use marginal likelihood  $p(\mathbf{X}|m)$  to select the best model from a finite set of models
 
$$\hat{m} = \arg \max_m p(m|\mathbf{X}) = \arg \max_m p(\mathbf{X}|m)p(m) = \arg \max_m p(\mathbf{X}|m), \text{ if } p(m) \text{ is uniform}$$
- Also useful for estimating hyperparam of the assumed model (if we consider  $m$  as the hyperparams)
  - Suppose hyperparams of likelihood are  $\alpha_\ell$  and that of prior are  $\alpha_p$  (so here  $m = \{\alpha_\ell, \alpha_p\}$ )
  - Assuming  $p(\alpha_\ell, \alpha_p)$  is uniform, hyperparams can be estimated via MLE-II (a.k.a. empirical Bayes)

$$\{\hat{\alpha}_\ell, \hat{\alpha}_p\} = \arg \max_{\alpha_\ell, \alpha_p} p(\mathbf{X}|\alpha_\ell, \alpha_p) = \arg \max_{\alpha_\ell, \alpha_p} \int p(\mathbf{X}|\boldsymbol{\theta}, \alpha_\ell)p(\boldsymbol{\theta}|\alpha_p)d\boldsymbol{\theta}$$

- Again, note that the integral here may be intractable and may need to be approximated
- Can also compute  $p(m|\mathbf{X})$  and do Bayesian Model Averaging :  $p(\mathbf{x}_*|\mathbf{X}) = \sum_{m=1}^M p(\mathbf{x}_*|\mathbf{X}, m)p(m|\mathbf{X})$



# Bayesian Inference for Multinoulli/Multinomial

- Assume  $N$  discrete-valued observations  $\{x_1, \dots, x_N\}$  with each  $x_n \in \{1, \dots, K\}$ , e.g.,
  - $x_n$  represents the outcome of a dice roll with  $K$  faces
  - $x_n$  represents the class label of the  $n$ -th example (total  $K$  classes)
  - $x_n$  represents the identity of the  $n$ -th word in a sequence of words
- Assume likelihood to be multinoulli with unknown params  $\pi = [\pi_1, \dots, \pi_K]$  s.t.  $\sum_{k=1}^K \pi_k = 1$

$$p(x_n | \pi) = \text{multinoulli}(x_n | \pi) = \prod_{k=1}^K \pi_k^{\mathbb{I}[x_n=k]}$$

- $\pi$  is a vector of probabilities (“probability vector”), e.g.,
  - Biases of the  $K$  sides of the dice
  - Prior class probabilities in multi-class classification
  - Probabilities of observing each words in the vocabulary
- Assume a **conjugate** Dirichlet prior on  $\pi$  with hyperparams  $\alpha = [\alpha_1, \dots, \alpha_K]$  (also,  $\alpha_k \geq 0, \forall k$ )

$$p(\pi | \alpha) = \text{Dirichlet}(\pi | \alpha_1, \dots, \alpha_K) = \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K \pi_k^{\alpha_k - 1} = \frac{1}{B(\alpha)} \prod_{k=1}^K \pi_k^{\alpha_k - 1}$$

$$\text{Beta Function: } B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt$$

# Bayesian Inference for Multinoulli/Multinomial

- The posterior over  $\pi$  is easy to compute in this case due to conjugacy b/w multinoulli and Dirichlet

$$p(\pi|\mathbf{X}, \alpha) = \frac{p(\mathbf{X}|\pi, \alpha)p(\pi|\alpha)}{p(\mathbf{X}|\alpha)} = \frac{p(\mathbf{X}|\pi, \alpha)p(\pi|\alpha)}{p(\mathbf{X}|\alpha)}$$

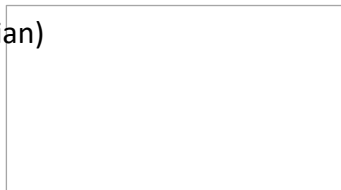
- Assuming  $x_n$ 's are i.i.d. given  $\pi$ ,  $p(\mathbf{X}|\pi) = \prod_{n=1}^N p(x_n|\pi)$ , therefore

$$p(\pi|\mathbf{X}, \alpha) \propto \prod_{n=1}^N \prod_{k=1}^K \pi_k^{\mathbb{I}[x_n=k]} \prod_{k=1}^K \pi_k^{\alpha_k-1} = \prod_{k=1}^K \pi_k^{\alpha_k + \sum_{n=1}^N \mathbb{I}[x_n=k] - 1}$$

- Even without computing the normalization constant  $p(\mathbf{X}, \alpha)$ , we can see that it's a Dirichlet! :-)
- Denoting  $N_k = \sum_{n=1}^N \mathbb{I}[x_n = k]$ , i.e., number of observations with value k, the posterior will be

$$p(\pi|\mathbf{X}, \alpha) = \text{Dirichlet}(\pi|\alpha_1 + N_1, \dots, \alpha_k + N_k)$$

- Note:  $N_1, \dots, N_k$  are the **sufficient statistics** in the this case
- Note: If we want, we can also get the MAP estimate of  $\pi$  (mode of the above Dirichlet)
  - MAP estimation via standard way will require solving a constraint opt. problem (via Lagrangian)



# Bayesian Inference for Multinoulli/Multinomial

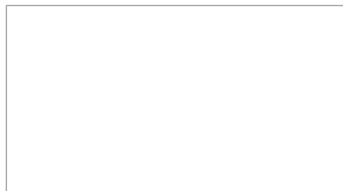
- Finally, let's also look at the **posterior predictive distribution** (i.e., the probability distribution of a new observation  $x_* \in \{1, \dots, K\}$  given the previous observation  $\mathbf{X} = \{x_1, \dots, x_N\}$ )

$$p(x_* | \mathbf{X}, \alpha) = \int p(x_* | \pi) p(\pi | \mathbf{X}, \alpha) d\pi$$

- Note that  $p(x_* | \pi) = \text{multinoulli}(x_* | \pi)$  and  $p(\pi | \mathbf{X}, \alpha) = \text{Dirichlet}(\pi | \alpha_1 + N_1, \dots, \alpha_K + N_K)$
- We can compute the posterior predictive for each possible outcome (K possibilities)

$$\begin{aligned} p(x_* = k | \mathbf{X}, \alpha) &= \int p(x_* = k | \pi) p(\pi | \mathbf{X}, \alpha) d\pi \\ &= \int \pi_k \times \text{Dirichlet}(\pi | \alpha_1 + N_1, \dots, \alpha_K + N_K) d\pi \\ &= \frac{\alpha_k + N_k}{\sum_{k=1}^K \alpha_k + N} \quad (\text{expectation of } \pi_k \text{ under the Dirichlet posterior}) \end{aligned}$$

- Therefore the posterior predictive distribution is multinoulli with posterior mean given as above
- Note that the predictive probabilities are smoothed (the effect of averaging over all possible  $\pi$ 's)



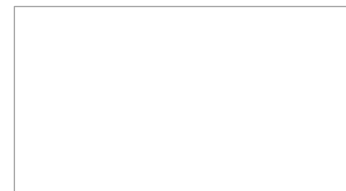
# Bayesian Inference for Mean of a Gaussian

- Consider  $N$  i.i.d. observations  $\mathbf{X} = \{x_1, \dots, x_N\}$  drawn from a one-dim Gaussian  $\mathcal{N}(x|\mu, \sigma^2)$

$$p(x_n|\mu, \sigma^2) = \mathcal{N}(x|\mu, \sigma^2) \propto \exp\left[-\frac{(x_n - \mu)^2}{2\sigma^2}\right]$$

$$p(\mathbf{X}|\mu, \sigma^2) = \prod_{n=1}^N p(x_n|\mu, \sigma^2)$$

- Assume the mean  $\mu \in \mathbb{R}$  for the Gaussian is unknown and assume variance  $\sigma^2$  to be known/fixed
- We wish to estimate the unknown  $\mu$  given the data  $\mathbf{X}$
- Let's do fully Bayesian inference for  $\mu$  (not MLE/MAP)
- We first need a prior distribution for the unknown param.  $\mu$
- Let's choose a Gaussian prior on  $\mu$ , i.e.,  $p(\mu) = \mathcal{N}(\mu|\mu_0, \sigma_0^2)$  with  $\mu_0, \sigma_0^2$  as fixed
- The prior basically says that the mean  $\mu$  is close to  $\mu_0$  (with some uncertainty depending on  $\sigma_0^2$ )



# Bayesian Inference for Mean of a Gaussian

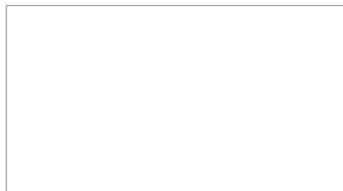
- The posterior distribution for the unknown mean parameter  $\mu$

$$p(\mu|\mathbf{X}) = \frac{p(\mathbf{X}|\mu)p(\mu)}{p(\mathbf{X})} \propto \prod_{n=1}^N \exp \left[ -\frac{(x_n - \mu)^2}{2\sigma^2} \right] \times \exp \left[ -\frac{(\mu - \mu_0)^2}{2\sigma_0^2} \right]$$

- Simplifying the above (using completing the squares trick) gives  $p(\mu|\mathbf{X}) \propto \exp \left[ -\frac{(\mu - \mu_N)^2}{2\sigma_N^2} \right]$  with

$$\frac{1}{\sigma_N^2} = \frac{1}{\sigma_0^2} + \frac{N}{\sigma^2}$$
$$\mu_N = \frac{\sigma^2}{N\sigma_0^2 + \sigma^2} \mu_0 + \frac{N\sigma_0^2}{N\sigma_0^2 + \sigma^2} \bar{x} \quad \left( \text{where } \bar{x} = \frac{\sum_{n=1}^N x_n}{N} \right)$$

- Posterior and prior have the same form (not surprising; the prior was conjugate to the likelihood)
- Consider what happens as  $N$  (number of observations) grows very large?
  - The posterior's variance  $\sigma_N^2$  approaches  $\sigma^2/N$  (and goes to 0 as  $N \rightarrow \infty$ )
  - The posterior's mean  $\mu_N$  approaches  $\bar{x}$  (which is also the MLE solution)





# Bayesian Inference for Mean of a Gaussian

- What is the **posterior predictive distribution**  $p(x_*|\mathbf{X})$  of a new observation  $x_*$ ?
- Using the inferred posterior  $p(\mu|\mathbf{X})$ , we can find the posterior predictive distribution

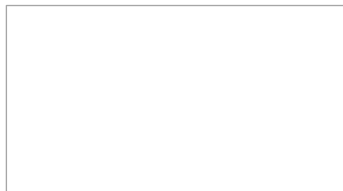
$$p(x_*|\mathbf{X}) = \int p(x_*|\mu, \sigma^2)p(\mu|\mathbf{X})d\mu = \int \mathcal{N}(x_*|\mu, \sigma^2)\mathcal{N}(\mu|\mu_N, \sigma_N^2)d\mu = \mathcal{N}(x_*|\mu_N, \sigma^2 + \sigma_N^2)$$

- Note; Can also get the above result by thinking of  $x_*$  as  $x_* = \mu + \epsilon$  where  $\mu \sim \mathcal{N}(\mu_N, \sigma_N^2)$ , and  $\epsilon \sim \mathcal{N}(0, \sigma^2)$  is independently added observation noise
- Note that, as per the above, the uncertainty in distribution of  $x_*$  now has two components
- $\sigma^2$ : Due to the noisy observation model,  $\sigma_N^2$ : Due to the uncertainty in  $\mu$
- In contrast, the **plug-in predictive posterior**, given a point estimate  $\hat{\mu}$  (e.g., MLE/MAP) would be

$$p(x_*|\mathbf{X}) = \int p(x_*|\mu, \sigma^2)p(\mu|\mathbf{X})d\mu \approx p(x_*|\hat{\mu}, \sigma^2) = \mathcal{N}(x_*|\hat{\mu}, \sigma^2)$$

... which doesn't incorporate the uncertainty in our estimate of  $\mu$  (since we used a point estimate)

- Note that as  $N \rightarrow \infty$ , both approaches would give the same  $p(x_*|\mathbf{X})$  since  $\sigma_N^2 \rightarrow 0$



# Bayesian Inference for Variance of a Gaussian

- Again consider N i.i.d. observations  $\mathbf{X} = \{x_1, \dots, x_N\}$  drawn from a one-dim Gaussian  $\mathcal{N}(x|\mu, \sigma^2)$

$$p(x_n|\mu, \sigma^2) = \mathcal{N}(x|\mu, \sigma^2) \text{ and } p(\mathbf{X}|\mu, \sigma^2) = \prod_{n=1}^N p(x_n|\mu, \sigma^2)$$

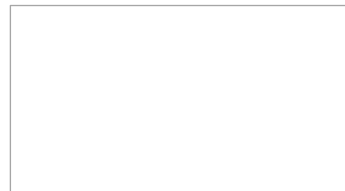
- Assume the variance  $\sigma^2 \in \mathbb{R}_+$  of the Gaussian is unknown and assume mean  $\mu$  to be known/fixed
- Let's estimate  $\sigma^2$  given the data  $\mathbf{X}$  using fully Bayesian inference (not MLE/MAP)
- We first need a prior distribution for  $\sigma^2$ . What prior  $p(\sigma^2)$  to choose in this case?
- If we want a conjugate prior, it should have the same form as the likelihood

$$p(x_n|\mu, \sigma^2) \propto (\sigma^2)^{-1/2} \exp\left[-\frac{(x_n - \mu)^2}{2\sigma^2}\right]$$

- An **inverse-gamma** prior  $\text{IG}(\alpha, \beta)$  has this form ( $\alpha, \beta$  are shape and scale hyperparams, resp.)

$$p(\sigma^2) \propto (\sigma^2)^{-(\alpha+1)} \exp\left[-\frac{\beta}{\sigma^2}\right] \quad (\text{note: mean of } \text{IG}(\alpha, \beta) = \frac{\beta}{\alpha-1})$$

- (Verify) The posterior  $p(\sigma^2|\mathbf{X}) = \text{IG}\left(\alpha + \frac{N}{2}, \beta + \frac{\sum_{n=1}^N (x_n - \mu)^2}{2}\right)$ . Again IG due to conjugacy.

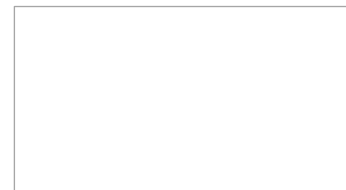


# Recap: Bayesian Generative Classification

- Recall generative classification  $p(y = k|x) = \frac{p(y=k)p(x|y=k)}{\sum_{k=1}^K p(y=k)p(x|y=k)}$ . Prediction rule for a test input  $\mathbf{x}_*$

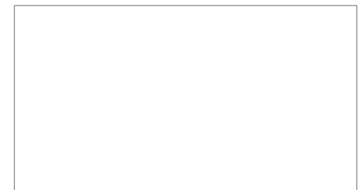
$$\begin{aligned} p(y_* = k|x_*, \mathbf{X}, \mathbf{y}) &= \frac{p(y_* = k|\mathbf{X}, \mathbf{y})p(x_*|y_* = k, \mathbf{X}, \mathbf{y})}{\sum_{k=1}^K p(y_* = k|\mathbf{X}, \mathbf{y})p(x_*|y_* = k, \mathbf{X}, \mathbf{y})} \\ &= \frac{p(y_* = k|\mathbf{y})p(x_*|\mathbf{X}^{(k)})}{\sum_{k=1}^K p(y_* = k|\mathbf{y})p(x_*|\mathbf{X}^{(k)})} \quad (1) \end{aligned}$$

- Note:  $\mathbf{X}^{(k)}$  denotes training inputs with  $y=k$
- Here  $p(y_* = k|\mathbf{y}) = \int p(y_*|\pi)p(\pi|\mathbf{y})d\pi$  (we did this; recall dice roll example)
- Here  $p(x_*|\mathbf{X}^{(k)}) = \int p(x_*|\theta_k)p(\theta_k|\mathbf{X}^{(k)})d\theta_k$  (post. Predictive dist. Of input  $\mathbf{x}_*$  under class  $k$ )
- Eq(1) is the posterior predictive distribution of test output  $y_*$  given input  $\mathbf{x}_*$ 
  - Note that we have done posterior averaging for all the parameters
- In contrast, for gen. class with MLE/MAP ,  $p(y_* = k|\mathbf{y}) \approx \pi_k$  and  $p(x_*|\mathbf{X}^{(k)}) \approx p(x_*|\theta_k)$



# Bayesian Inference

- Bayesian Modelling
- Priors
- MLE and MAP
- Bayesian Inference Examples
- **Tractability and Approximation Methods**
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# Bayesian Modelling

Everything follows from two simple rules:

**Sum rule:**  $P(x) = \sum_y P(x, y)$

**Product rule:**  $P(x, y) = P(x)P(y|x)$

$$P(\theta|\mathcal{D}, m) = \frac{P(\theta|\mathcal{D}, m)P(\theta|m)}{P(\mathcal{D}|m)}$$

$P(\mathcal{D}|\theta, m)$  likelihood of parameters  $\theta$  in model  $m$

$P(\theta|m)$  prior probability of  $\theta$

$P(\theta|\mathcal{D}, m)$  posterior of  $\theta$  given data  $\mathcal{D}$

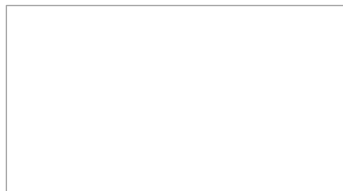
## Prediction:

$$P(x|\mathcal{D}, m) = \int P(x|\theta, \mathcal{D}, m)P(\theta|\mathcal{D}, m)d\theta$$

## Model Comparison:

$$P(m|\mathcal{D}) = \frac{P(\mathcal{D}|m)P(m)}{P(\mathcal{D})}$$

$$P(\mathcal{D}|m) = \int P(\mathcal{D}|\theta, m)P(\theta|m)d\theta$$



# Computing Marginal Likelihoods can be Computationally Intractable

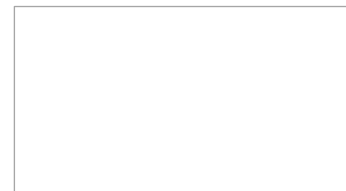
Observed data  $x$ , hidden variables  $z$ , parameters  $\theta$ , model class  $m$ .

$$p(x|m) = \int p(x|\theta, m)p(\theta|m)d\theta$$

- This can be a very **high dimensional integral**.
- The presence of **latent variables** results in additional; dimensions that need to be marginalized out.

$$p(x|m) = \iint p(x, z | \theta, m)p(\theta|m)dz d\theta$$

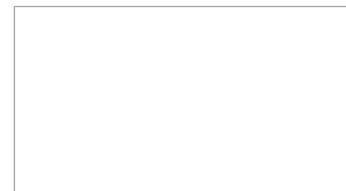
- The likelihood term can be **complicated**.



# Approximation Methods for Posteriors and Marginal Likelihoods

- Laplace approximation
- Bayesian Information Criterion (BIC)
- Variational approximations
- Expectation Propagation (EP)
- ...

Note: there are many other deterministic approximations; we won't review them in this lecture.



# Laplace Approximation

Data set  $x$ , modes  $m, m', \dots$  parameter  $\theta, \theta' \dots$

Model Comparison:  $p(m|x) \propto p(m)$   $p(x|m)$

For large amounts of data (relative to number of parameters,  $d$ ) the parameter posterior is approximately

Gaussian around the MAP estimate  $\hat{\theta}$ :

$$p(\theta|x, m) \approx (2\pi)^{-\frac{d}{2}} |A|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}(\theta - \hat{\theta})^T A(\theta - \hat{\theta})\right\}$$

Where  $-A$  is the  $d \times d$  Hessian of the log posterior  $A_{ij} = -\frac{d^2}{d\theta_i d\theta_j} \ln p(\theta|x, m)|_{\theta=\hat{\theta}}$

$$p(x|m) = \frac{p(\theta, x|m)}{p(\theta|x, m)}$$

Evaluating the above expression for  $\ln p(x|m)$  at  $\hat{\theta}$ :

$$\ln p(x|m) \approx \ln p(\hat{\theta}|m) + \ln p(x|\hat{\theta}, m) + \frac{d}{2} \ln 2\pi - \frac{1}{2} \ln |A|$$

This can be used for model comparison / selection.



# Bayesian Information Criterion (BIC)

BIC can be obtained from the Laplace approximation:

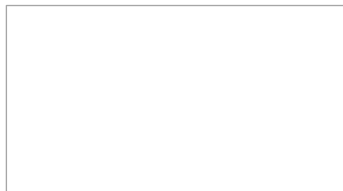
$$\ln p(x|m) \approx \ln p(\hat{\theta}|m) + \ln p(x|\hat{\theta}, m) + \frac{d}{2} \ln 2\pi - \frac{1}{2} \ln |A|$$

By taking the large sample limit ( $n \rightarrow \infty$ ) where  $n$  is the number of data points:

$$\ln p(x|m) \approx \ln p(x|\hat{\theta}, m) - \frac{d}{2} \ln n$$

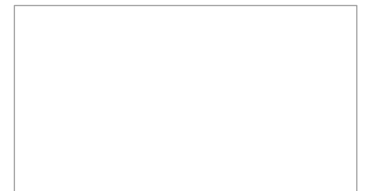
## Properties:

- Quick and easy to compute
- It does not depend on the prior
- We can use the ML estimate of  $\theta$  instead of the MAP estimate
- It is equivalent to the MDL criterion
- Assumes that as  $n \rightarrow \infty$ , all the parameters are well-determined (i.e. the model is **identifiable**; otherwise,  $d$  should be the number of **well-determined** parameters)
- **Danger:** counting parameters can be deceiving! (c.f. sinusoid, infinite models)



# Bayesian Inference

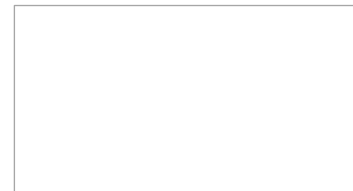
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# Goals of Inference

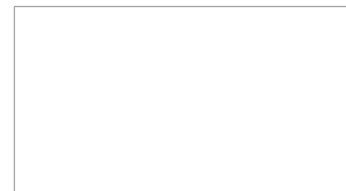
**Recall, the goals of inference in graphical models include:**

- Computing the likelihood of observed data (in models with latent variables).
- Computing the marginal distribution over a given subset of nodes in the model.
- Computing the conditional distribution over a given subset of nodes given a disjoint subset of nodes.
- Computing a mode of the density (for the above distributions).



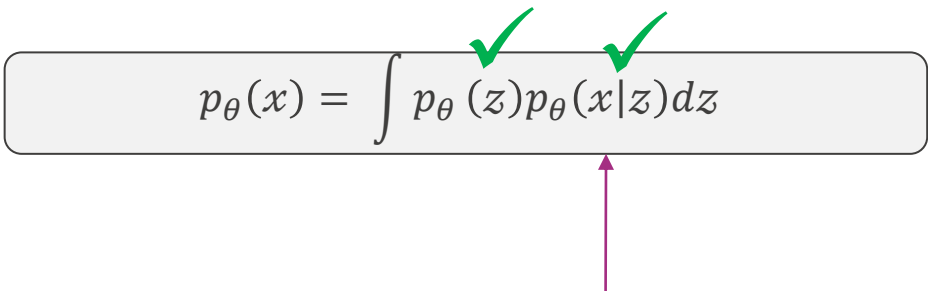
# Approaches to Inference

- **Exact inference algorithms:**
  - Brute force.
  - The elimination algorithm.
  - Message passing (sum-product algorithm, belief propagation).
  - Junction tree algorithm.
- **Approximate inference algorithms:**
  - Loopy belief propagation
  - **Variational (Bayesian) inference + mean field approximations**
  - Stochastic simulation / sampling / MCMC

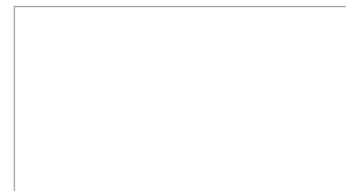


# Posterior Inference: Intractability

Data likelihood:

$$p_{\theta}(x) = \int p_{\theta}(z) p_{\theta}(x|z) dz$$
The equation is enclosed in a light gray rounded rectangle. Two green checkmarks are positioned above the terms  $p_{\theta}(z)$  and  $p_{\theta}(x|z)$ . A purple arrow points upwards from the text below to the  $p_{\theta}(x|z)$  term.

Intractable to compute  
 $p(x|z)$  for every  $z$ !



# Posterior Inference: Intractability

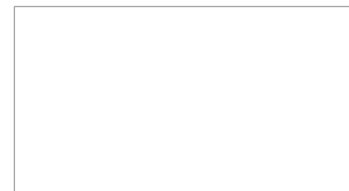
Data likelihood:

$$p_{\theta}(x) = \int p_{\theta}(z) p_{\theta}(x|z) dz$$

Posterior density also intractable:

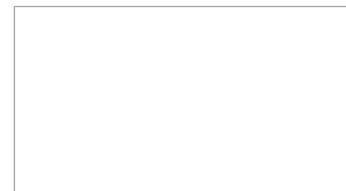
$$p_{\theta}(z|x) = p_{\theta}(x|z) p_{\theta}(z) / p_{\theta}(x)$$

Intractable data likelihood



Approximate intractable posterior  $p(z|x)$  with a manageable distribution  $q(z)$

Minimize the KL divergence:  $D_{KL}(q(z) \parallel p(z|x))$



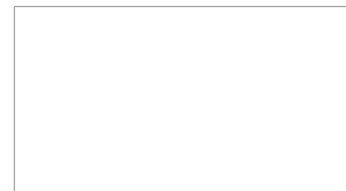
# Motivating Example

## Bayesian mixture of Gaussians

The likelihood (i.e. the generative process)

1. Draw  $\mu_k \sim \mathcal{N}(0, \mathcal{I}^2)$  for  $k=1, \dots, K$
2. For  $i=1, \dots, n$ 
  - a. Draw  $z_i \sim \text{Cat}(\pi)$
  - b. Draw  $x_i \sim \mathcal{N}(\mu_{z_i}, \sigma^2)$

Note that we have observed variables,  $x_{1:n}$ , latent variables  $\mu_{1:K}$  and  $z_{1:n}$ , and parameters  $\{\mathcal{I}^2, \pi, \sigma^2\}$



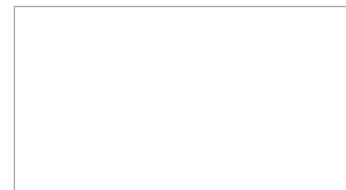


# Motivating Example

- We can write the posterior distribution as:

$$p(\mu_{1:K}, z_{1:n} | x_{1:n}) = \frac{\prod_{k=1}^K (\mu_k) \prod_{i=1}^n p(z_i) p(x_i | z_i, \mu_{1:K})}{\int_{\mu_{1:K}} \sum_{z_{1:n}} \prod_{k=1}^K p(\mu_k) \prod_{i=1}^n p(z_i) p(x_i | z_i, \mu_{1:K}) d\mu}$$

- Where we have suppressed writing the parameters for ease of notation



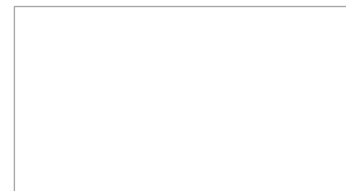
# Motivating Example

- Can we compute this density?
- The numerator can be computed for any choice of the latent variables
- The problem is the denominator (the marginal probability of the observations)

$$p(x_{1:n}) = \int_{\mu_{1:K}} \sum_{z_{1:n}} \prod_{k=1}^K p(\mu_k) \prod_{i=1}^n p(z_i) p(x_i | z_i, \mu_{1:K}) d\mu$$

$$= \int_{\mu_{1:K}} \prod_{k=1}^K p(\mu_k) \prod_{i=1}^n \sum_{z_{1:n}} p(z_i) p(x_i | z_i, \mu_{1:K}) d\mu$$

- This integral cannot easily be computed analytically



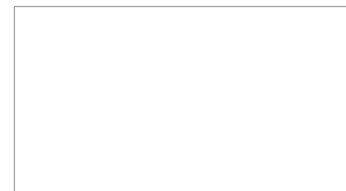
# Variational Bayes

## The main idea behind variational Bayes:

- Choose a family of distributions over the latent variables  $z_{1:m}$  with its own set of variational parameters  $v$ , i.e.

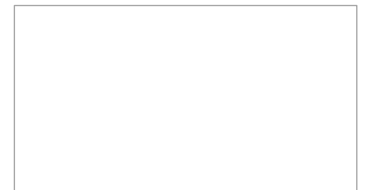
$$q(z_{1:m}|v)$$

- Then, we find the setting of the parameters that makes our approximation  $q$  closed to the posterior distribution
  - This is where optimization algorithms come in
- Then we can use  $q$  with the fitted parameters in places of the posterior
  - e.g., to form predictions about future data, or to investigate the posterior distribution over the hidden variables, find modes, etc.



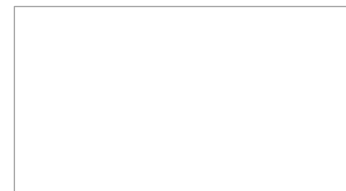
# Bayesian Inference

- Bayesian Modelling
- Priors
- MLE and MAP
- Bayesian Inference Examples
- Tractability and Approximation Methods
- Variational Inference
- **Evidence Lower Bound (ELBO)**
- Kullback-Liebler (KL) Divergence
- Mean field Variational Inference
- Variational Bayes Learning
- Myths about Bayesian Methods



# The Evidence Lower Bound

- So , to do variational Bayes, we want to minimize the KL divergence between our approximation  $q$  and our posterior  $p$ .
- However, we cant actually minize this quantity (we will show why later), but we can minimize a function that is qual to it up to a constant
- This function is known as the **evidence lower bound (ELBO)**
- Recall that the “evidence” is a term used for the marginal likelihood of observations (or the log of the)

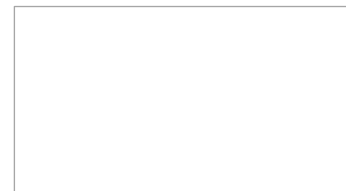


# Deriving the Evidence Lower Bound

- First recall Jensen's inequality (applied to random variables  $X$ ) :  
When  $f$  is concave,  $f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$ .
- We apply Jensen's inequality to the log (marginal) probability of the observations to get the ELBO.

$$\begin{aligned}
 \log p(x) &= \log \int_{\mathcal{Z}} p(x, z) dz \\
 &= \log \int_{\mathcal{Z}} p(x, z) \frac{q(z)}{q(z)} dz \\
 &= \log \left( \mathbb{E}_q \left[ \frac{p(x, z)}{q(z)} \right] \right) \\
 &\geq \mathbb{E}_q [\log p(x, z)] - \mathbb{E}_q [\log q(z)]
 \end{aligned}$$

This final line is the ELBO! It is a lower bound for the evidence.

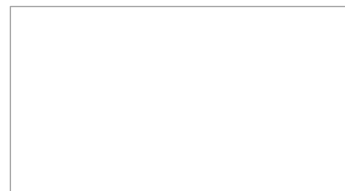


# The Evidence Lower Bound

All together, the Evidence Lower Bound (ELBO) for a probability model  $p(x, z)$  and approximation  $q(z)$  to the posterior is :

$$\mathbb{E}_q[\log p(x, z)] - \mathbb{E}_q[\log q(z)]$$

- This quantity is less than or equal to the evidence (log marginal probability of the observations).
- We optimize this quantity (over densities  $q(z)$ ) in Variational Bayes to find an “optimal approximation”.



# The Evidence Lower Bound

## Why do we maximize the ELBO?

- First recall that

$$p(z|x) = \frac{p(z, x)}{p(x)}$$

- Next, we can write the KL divergence as:

$$\begin{aligned}\text{KL}(q \parallel p) &= \mathbb{E}_q \left[ \log \frac{q(z)}{p(z|x)} \right] \\ &= \mathbb{E}_q [\log q(z)] - \mathbb{E}_q [\log p(z|x)] \\ &= \mathbb{E}_q [\log q(z)] - \mathbb{E}_q [\log p(z, x) + \log p(x)] \\ &= -(\mathbb{E}_q [\log p(z, x)] - \mathbb{E}_q [\log q(z)]) + \log p(x)\end{aligned}$$

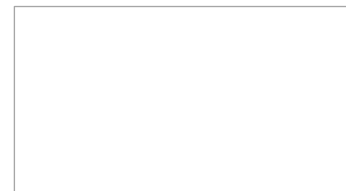
This final line is the negative ELBO plus a constant (that does not depend on  $q$ ).



# The Evidence Lower Bound

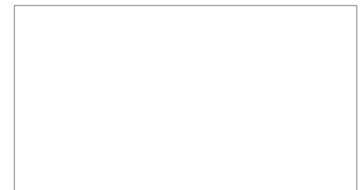
Hence...

- Therefore, finding an approximation  $q$  that maximizes the ELBO is equivalent to finding the  $q$  that minimizes the KL divergence to the posterior!
- Note: The difference between the ELBO and the KL divergence is the log normalizer (i.e. the evidence), which is the quantity that the ELBO bounds.



# Bayesian Inference

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# Kullback-Leibler divergence

$$D_{KL}(q \parallel p) = \int q(z) \log \frac{q(z)}{p(z)} dz$$

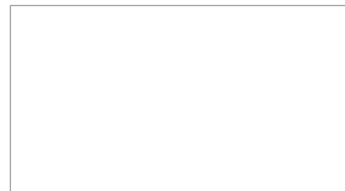
$$\begin{aligned} D_{KL}(q \parallel p) &= - \int q(z) \log \left( \frac{p(z)}{q(z)} \right) dz \\ &= E_q \left[ -\log \left( \frac{p}{q} \right) \right] \end{aligned}$$

- A result from information theory : For any  $P$  and  $Q$

$$KL(P \parallel Q) \geq 0$$

- $KL(P \parallel Q) = 0$  if any if  $P \equiv Q$
- $D$  is asymmetric

The average information gained from moving from  $q$  to  $p$



# Kullback-Leibler divergence

We measure the closeness of the two distributions with the Kullback-Leibler (KL) divergence, defined to be

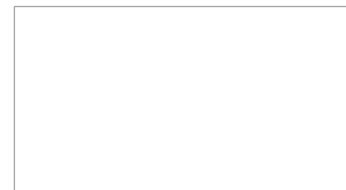
$$\text{KL}(q \parallel p) = \int_{\mathcal{Z}} q(z) \log \frac{q(z)}{p(z|x)} dz = \mathbb{E}_q \left[ \log \frac{q(z)}{p(z|x)} \right]$$

**Intuitively, there are three “cases” of importance:**

- If  $q$  is high and  $p$  is high, then we are happy (i.e. low KL divergence).
- If  $q$  is high and  $p$  is low then we pay a price (i.e. high KL divergence).
- If  $q$  is low then we don't care (i.e. also low KL divergences, regardless of  $p$ ).

**Intuitively, it might make more sense to consider  $\text{KL}(p \parallel q)$**

- However, we do not do this for computational reasons (which we will explain).



# M-projection vs. I-projection

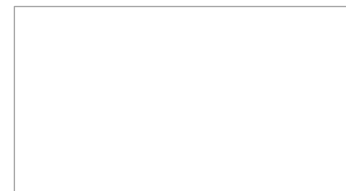
- M-projection of  $Q$  onto  $P$

$$Q^* = \operatorname{argmin}_{Q \in \mathcal{Q}} KL(P || Q)$$

- I-projection of  $Q$  onto  $P$

$$Q^* = \operatorname{argmin}_{Q \in \mathcal{Q}} KL(Q || P)$$

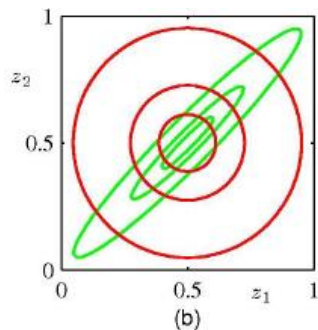
- These two will differ only when  $Q$  is minimized over a restricted set of probability distributions (when  $P \notin \mathcal{Q}$  set of possible  $Q$  distributions)



# KL divergence: M-projection vs. I-projection

- Let  $P$  be a **2D Gaussian** and  $Q$  be a **Gaussian distribution with diagonal covariance matrix**:

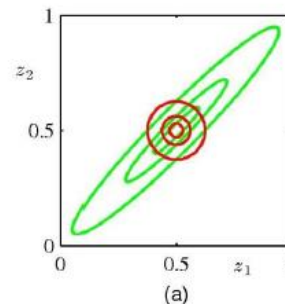
$$Q^* = \operatorname{argmin}_Q \int P(\mathbf{z}) \log \frac{P(\mathbf{z})}{Q(\mathbf{z})} d\mathbf{z}$$



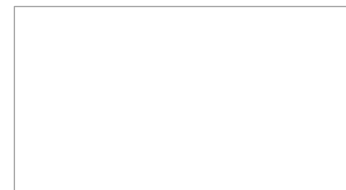
$$E_P[\mathbf{z}] = E_Q[\mathbf{z}]$$

$P$ : Green  
 $Q^*$ : Red

$$Q^* = \operatorname{argmin}_Q \int Q(\mathbf{z}) \log \frac{Q(\mathbf{z})}{P(\mathbf{z})} d\mathbf{z}$$



$$E_P[\mathbf{z}] = E_Q[\mathbf{z}]$$

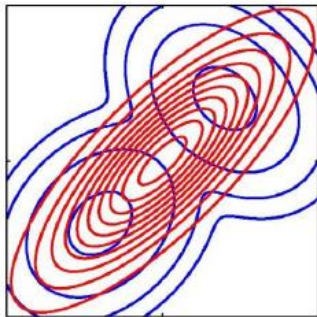


# KL divergence: M-projection vs. I-projection

- Let  $P$  a mixture of two **2D Gaussian** and  $Q$  be a **2D Gaussian** distribution with arbitrary covariance matrix:

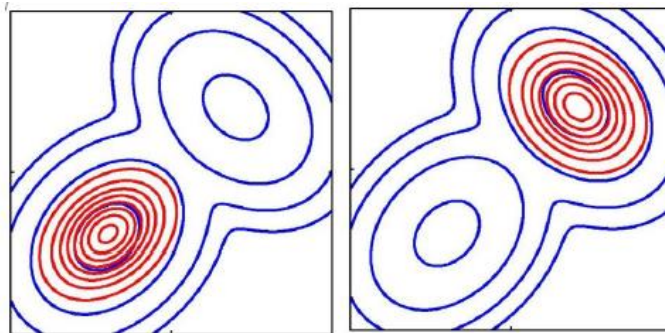
$$Q^* = \operatorname{argmin}_Q \int P(\mathbf{z}) \log \frac{P(\mathbf{z})}{Q(\mathbf{z})} d\mathbf{z}$$

$P$ : Blue  
 $Q^*$ : Red

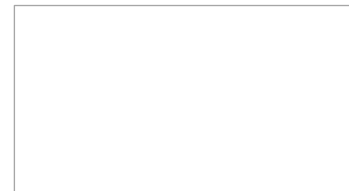


$$\begin{aligned} E_P[\mathbf{z}] &= E_Q[\mathbf{z}] \\ \operatorname{cov}_P[\mathbf{z}] &= \operatorname{cov}_Q[\mathbf{z}] \end{aligned}$$

$$Q^* = \operatorname{argmin}_Q \int Q(\mathbf{z}) \log \frac{Q(\mathbf{z})}{P(\mathbf{z})} d\mathbf{z}$$



**Two good solutions!**



## M-projection

- Computing  $KL(P||Q)$  requires inference on P

$$KL(P||Q) = \sum_z P(z) \log \frac{P(z)}{Q(z)} = -H[P] - E_P[\log Q(z)]$$



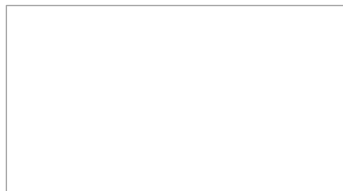
Inference on P (that is difficult) is required!

- When  $Q$  is in the exponential family:

$$KL(P||Q) = 0 \Leftrightarrow E_P[T(z)] = E_Q[T(z)]$$

Moment projection

- Expectation Propagation methods are based on minimizing  $KL(P||Q)$



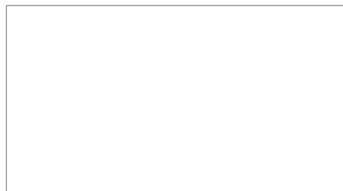


## I-projection

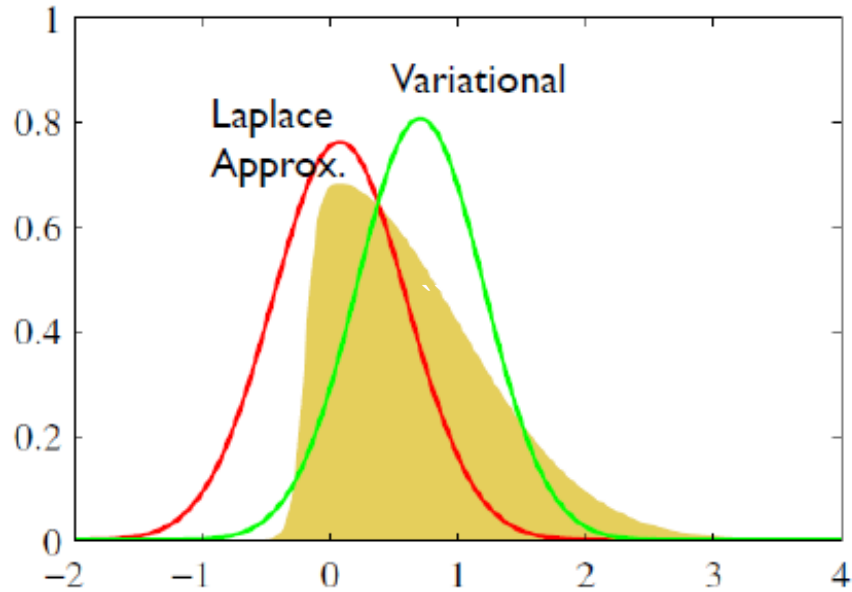
- $KL(Q||P)$  can be computed without performing inference on  $P$

$$KL(Q||P) = \int Q(z) \log \frac{Q(z)}{P(z)} dz = -H[Q] - E_q[\log P(z)]$$

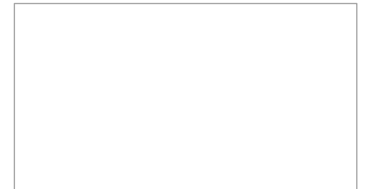
- Most variational inference algorithms make use of  $KL(Q||P)$
- Computing expectations w.r.t.  $Q$  is tractable (by choosing a suitable class of distributions for  $Q$ )
  - We choose a restricted family of distributions such that the expectations can be evaluated and optimized efficiently.
  - And yet which is still sufficiently flexible as to give a good approximation



# Example of variational approximation



[Bishop]



# KL-divergence as an objective function

It's well-known that KL-divergence is not symmetric, but which direction is right for fitting your model?

Which KL is which?

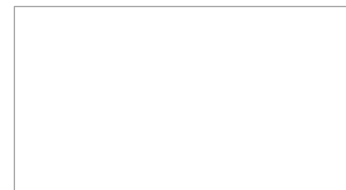
If we're fitting  $q_\theta$  to  $p$  using

**KL( $p \parallel q_\theta$ )**

- Mean seeking, inclusive (more principled because approximates the full distribution)
- Requires normalization wrt  $p$  (i.e. often not computationally convenient)

**KL( $q_\theta \parallel p$ )**

- Mode seeking, exclusive
- No normalization wrt  $p$  (i.e. computationally convenient)

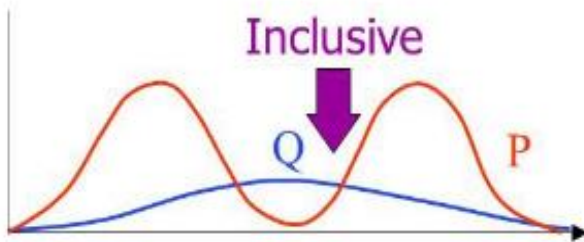
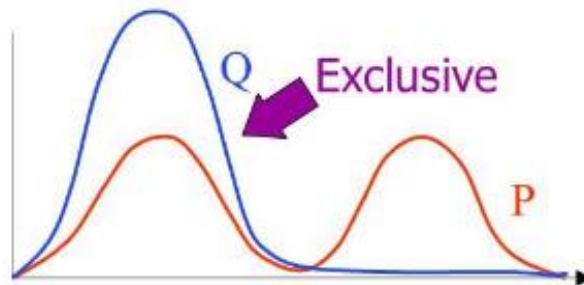


# KL-divergence as an objective function

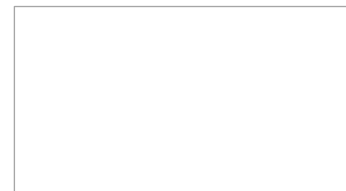
Inclusive vs. exclusive divergence

$$\begin{aligned} &\text{Minimising} \\ &\mathbf{KL}(\mathbf{Q} \parallel \mathbf{P}) \\ &= \sum_H Q(H) \ln \frac{Q(H)}{P(H|V)} \end{aligned}$$

$$\begin{aligned} &\text{Minimising} \\ &\mathbf{KL}(\mathbf{P} \parallel \mathbf{Q}) \\ &= \sum_H P(H|V) \ln \frac{P(H|V)}{Q(H)} \end{aligned}$$



<https://timvieira.github.io/blog/post/2014/10/06/kl-divergence-as-an-objective-function/>



# KL-divergence as an objective function

## Computational perspective

Let's look at what's involved in fitting a model  $q_\theta$  in each direction. In this section, I'll describe the gradient and pay special attention to the issue of normalization.

**Notation:**  $p, q_\theta$  are probability distributions,  $p = \bar{p}/Z_p$ , where  $Z_p$  is the normalization constant.

The easy direction  $\text{KL}(q_\theta \parallel p)$

$$\begin{aligned}\text{KL}(q_\theta \parallel p) &= \sum_d q(d) \log \left( \frac{q(d)}{p(d)} \right) \\ &= \sum_d q(d) (\log q(d) - \log p(d)) \\ &= \underbrace{\sum_d q(d) \log q(d)}_{\text{entropy}} - \underbrace{\sum_d q(d) \log p(d)}_{\text{Cross-entropy}}\end{aligned}$$

Let's look at normalization of  $p$ , the entropy term is easy because there is no  $p$  in it.

$$\begin{aligned}\sum_d q(d) \log p(d) &= \sum_d q(d) \log(\bar{p}(d)/Z_p) \\ &= \sum_d q(d) ((\log \bar{p}(d) - \log Z_p)) \\ &= \sum_d q(d) \log \bar{p}(d) - \sum_d q(d) \log Z_p \\ &= \sum_d q(d) \log \bar{p}(d) - \log Z_p\end{aligned}$$

# KL-divergence as an objective function

In this case,  $-\log \mathbf{Z}_p$  is an additive constant, which can be dropped because we're optimizing.

This leaves us with the following optimization problem:

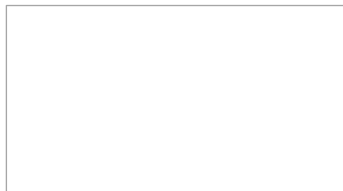
$$\operatorname{argmin}_{\theta} \mathbf{KL}(q_{\theta} \parallel p) = \operatorname{argmin}_{\theta} \sum_d q_{\theta}(d) \log q_{\theta}(d) - \sum_d q_{\theta}(d) \log \bar{p}(d)$$

Let's work out the gradient

$$\begin{aligned} \nabla \left[ \sum_d q_{\theta}(d) \log q_{\theta}(d) - \sum_d q_{\theta}(d) \log \bar{p}(d) \right] \\ &= \sum_d \nabla [q_{\theta}(d) \log q_{\theta}(d)] - \sum_d \nabla [q_{\theta}(d)] \log \bar{p}(d) \\ &= \sum_d \nabla [q_{\theta}(d)] (1 + \log q_{\theta}(d) - \log \bar{p}(d)) \\ &= \sum_d \nabla [q_{\theta}(d)] (\log q_{\theta}(d) - \log \bar{p}(d)) \end{aligned}$$

We killed the one in the last equality because  $\sum_d \nabla [\sum_d \mathbf{q}(d)] = \nabla [\mathbf{1}] = \mathbf{0}$ , for any  $\mathbf{q}$  which is a probability distribution.

This direction is convenient because we don't need to normalize  $p$ . Unfortunately, the "easy" direction is nonconvex in general – unlike the "hard" direction, which (as we'll see shortly) is convex.



# KL-divergence as an objective function

**Harder direction :  $\text{KL}(\mathbf{p} \parallel \mathbf{q}_\theta)$**

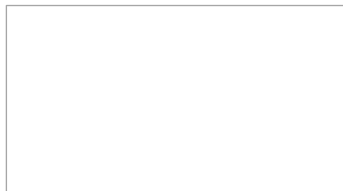
$$\begin{aligned}\text{KL}(\mathbf{p} \parallel \mathbf{q}_\theta) &= \sum_d p(d) \log \left( \frac{p(d)}{q(d)} \right) \\ &= \sum_d p(d) (\log p(d) - \log q(d)) \\ &= \sum_d p(d) \log p(d) - \sum_d p(d) \log q(d)\end{aligned}$$

Clearly the first term (entropy) won't matter if we're just trying to optimize wrt  $\theta$ . So, let's focus on the second term (cross-entropy).

$$\begin{aligned}\sum_d p(d) \log q(d) &= \frac{1}{Z_p} \sum_d \bar{p}(d) \log(\bar{p}(d)/Z_q) \\ &= \frac{1}{Z_p} \sum_d \bar{p}(d) (\log \bar{q}(d) - \log Z_q) \\ &= \left( \frac{1}{Z_p} \sum_d \bar{p}(d) \log \bar{q}(d) \right) - \left( \frac{1}{Z_p} \sum_d \bar{p}(d) \log Z_q \right) \\ &= \left( \frac{1}{Z_p} \sum_d \bar{p}(d) \log \bar{q}(d) \right) - (\log Z_p) \left( \frac{1}{Z_p} \sum_d \bar{p}(d) \right) \\ &= \left( \frac{1}{Z_p} \sum_d \bar{p}(d) \log \bar{q}(d) \right) - \log Z_q\end{aligned}$$

The gradient, when  $q$  is the exponential family, is intuitive:

$$\begin{aligned}\nabla \left[ \frac{1}{Z_p} \sum_d \bar{p}(d) \log \bar{q}(d) - \log Z_q \right] &= \frac{1}{Z_p} \sum_d \bar{p}(d) \nabla [\log \bar{q}(d)] - \nabla \log Z_q \\ &= \frac{1}{Z_p} \sum_d \bar{p}(d) \phi_q(d) - \mathbb{E}_q[\phi_q] \\ &= \mathbb{E}_p[\phi_q] - \mathbb{E}_q[\phi_q]\end{aligned}$$



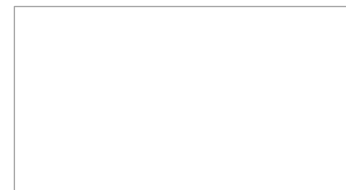
# KL-divergence as an objective function

Why do we say this hard to compute? Well, for most interesting models, we can't compute

$Z_p = \sum_d \bar{p}(d)$ . This is because  $p$  is presumed to be a complex model (e.g., the real world, an intricate factor graph, a complicated Bayesian posterior). If we can't compute  $Z_p$ , it's highly unlikely that we can compute another (nontrivial) integral under  $\bar{p}$ , e.g.,  $\sum_d \bar{p}(d) \log \bar{q}(d)$ .

Nonetheless, optimizing KL in this direction is still useful. Examples include: expectation propagation, variational decoding, and maximum likelihood estimation.  $p$  is the empirical distribution, so technically you don't have to compute its normalizing constant, but you do need samples from it, which can be just as hard to get as computing a normalization constant.

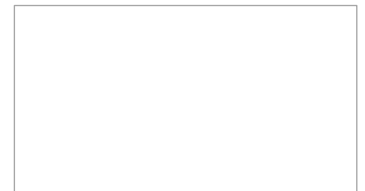
Optimization problem is convex when  $q_\theta$  is an exponential family – i.e., for any  $p$  the optimization problem is “easy”. You can think of maximum likelihood estimation (MLE) as a method which minimizes **KL**, divergence based on sample of  $p$ . In this case,  $p$  is the true data distribution! The first term in the gradient is based on a sample instead of an exact estimate (often called “observed feature counts”). The downside, of that computing  $\mathbb{E}_p[\phi_q]$  might not be tractable or, for MLE, require tons of sample.





# Bayesian Inference

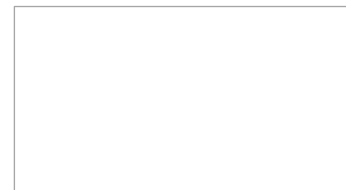
- Bayesian Modelling
- Priors
- MLE and MAP
- Bayesian Inference Examples
- Tractability and Approximation Methods
- Variational Inference
- Evidence Lower Bound (ELBO)
- Kullback-Liebler (KL) Divergence
- **Mean field Variational Inference**
- Variational Bayes Learning
- Myths about Bayesian Methods



## Quick Recap

### Quick recap on what we've covered so far:

- We often cannot compute posteriors, and so we need to approximate them, using (for e.g.) variational methods.
- In variational Bayes, we'd like to find an approximation within some family that minimizes the **KL** divergence to the posterior, but we can't directly minimize this.
- Therefore, we defined the ELBO, which we can maximize, and this is equivalent to minimizing the **KL** divergence.
- Next, we will discuss a specific family of approximations.



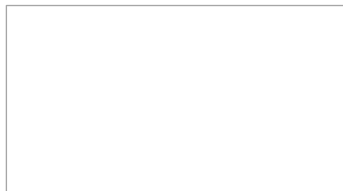
# Mean Field Variational Inference

- We now describe a popular family of variational approximations called **mean field approximations**.
- In this type of variational Inference, we assume the variational distribution over the latent variables factories as

$$q(z_1, \dots, z_m) = \prod_{j=1}^m q(z_j)$$

(where we omit variational parameters for ease of notation).

- **We refer to  $q(z_j)$ , the variational approximation for a single latent variable, as a “local variational approximation”.**
- In the above expression, the variational approximations  $q(z_j)$  over each latent variable  $z_j$  is independent.



# Mean Field Variational Inference

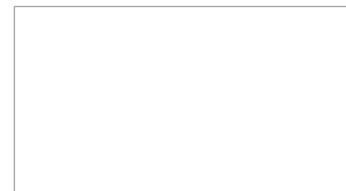
Note that this is a fairly general setup; we can also partition the latent variables  $z_1, \dots, z_m$  into  $R$  groups  $z_{G_1}, \dots, z_{G_R}$ , and use the approximation:

$$q(z_1, \dots, z_m) = q(z_{G_1}, \dots, z_{G_R}) = \prod_{r=1}^R q(z_{G_r})$$

- Often called “generalized mean field” versus (the previous) “naïve mean field”.
- More on this later, applied to Markov random fields.

**Typically, this approximation does not contain the true posterior (because the latent variables are dependent).**

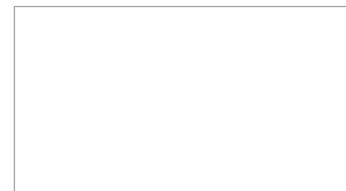
- E.g. In the (Bayesian) mixture of Gaussians model, all the cluster assignments  $z_i$  for  $i = 1, \dots, n$  are dependent on each other and on the cluster locations  $\mu_{1:K}$ , given data



# Optimizing the ELBO in Mean Field Variational Inference

**How do we optimize the ELBO in mean field variational inference?**

- Typically, we use coordinate ascent optimization.
- I.e. we optimize each latent variable's variational approximation  $q(z_j)$  in turn while holding the others fixed.
  - **At each iteration we get an updated “local” variational approximation.**
  - **And we iterate through each latent variable until convergence.**
- Note: this is not the only way to optimize the ELBO in mean field approximations (e.g. one can do gradient ascent, using the “natural gradient”), however it is a very popular method.



# Optimizing the ELBO in Mean Field Variational Inference

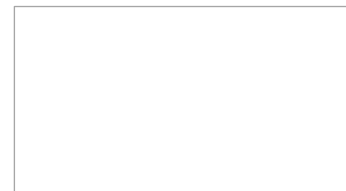
First, recall that the (probability) chain rule gives:

$$p(z_{1:m}, x_{1:n}) = p(x_{1:n}) \prod_{j=1}^m p(z_j | z_{1:(j-1)}, x_{1:n})$$

Note that the latent variables in this product can occur in any order (i.e. the indexing from 1 to m is arbitrary) --- this will be important later.

Second, note that we can decompose the entropy term of the ELBO (using the mean field variational approximation) as

$$\mathbb{E}_q[\log q(z_{1:m})] = \sum_{j=1}^m \mathbb{E}_{q_j}[\log q(z_j)]$$



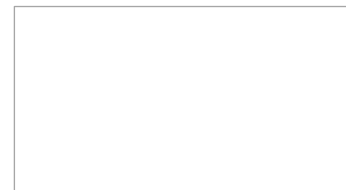
# Optimizing the ELBO in Mean Field Variational Inference

- Third, using the previous two facts, we can decompose the ELBO  $\mathcal{L}$  for the mean field variational approximation into a nice form.
- Recall that the ELBO is defined as:

$$\mathbb{E}_q[\log p(x, z)] - \mathbb{E}_q[\log q(z)]$$

- Therefore, under the mean field approximation, the ELBO can be written:

$$\mathcal{L} = \log p(x_{1:n}) + \sum_{j=1}^m \left( \mathbb{E}_q[\log p(z_j | z_{1:(j-1)}, x_{1:n})] - \mathbb{E}_{q_j}[\log q(z_j)] \right)$$



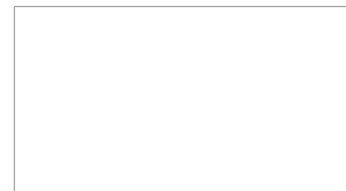
# Optimizing the ELBO in Mean Field Variational Inference

Before we can continue, we need to introduce some terminology:

- “The conditional” for latent variable  $z_j$  is:

$$p(z_j | z_1, \dots, z_{j-1}, z_{j+1}, \dots, z_m, x) = p(z_j | z_{-j}, x)$$

- Where the  $-j$  notation denotes all indices other than the  $j^{th}$ .
- This is actually the “posterior conditional” of  $z_j$ , given all other latent variables and observations.
- This posterior conditional is very important in mean field variational Bayes, and will be important in future inference algorithm used in this class, such as Gibbs sampling.





# Optimizing the ELBO in Mean Field Variational Inference

- Again, we wrote the ELBO  $\mathcal{L}$  for the mean field variational approximation as:

$$\mathcal{L} = \log p(x_{1:n}) + \sum_{j=1}^m \left( \mathbb{E}_q [\log p(z_j | z_{1:(j-1)}, x_{1:n})] - \mathbb{E}_{q_j} [\log q(z_j)] \right)$$

- Next, we want to derive the coordinate ascent update for a latent variable  $z_j$ , keeping all other latent variables fixed.
  - i.e. we want the  $\text{argmax}_{q_j} \mathcal{L}$ .
- Removing the parts that do not depend on  $q(z_j)$ , we can write:

$$\begin{aligned} \text{argmax}_{q_j} \mathcal{L} &= \text{argmax}_{q_j} \left( \mathbb{E}_q [\log p(z_j | z_{1:(j-1)}, x)] - \mathbb{E}_{q_j} [\log q(z_j)] \right) \\ &= \text{argmax}_{q_j} \left( \int q(z_j) \mathbb{E}_{q_{-j}} [\log p(z_j | z_{-j}, x)] dz_j - \int q(z_j) \log q(z_j) dz_j \right) \end{aligned}$$

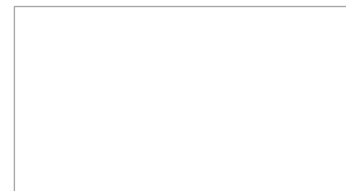
# Optimizing the ELBO in Mean Field Variational Inference

## Notes:

- In the previous expression, to get the term  $\mathbb{E}_q[\log p(\mathbf{z}_j | \mathbf{z}_{-j}, \mathbf{x})]$ , we have re-ordered the latent variables in our sum so that the  $j^{th}$  latent variable comes last.
- The notation  $\mathbb{E}_{q_{-j}}$  is the expectation over all “other” latent variables (except for the  $j^{th}$ ).
- We define the term inside the argmax on the last line to be called  $\mathcal{L}_j$ , i.e.

$$\mathcal{L}_j = \int q(\mathbf{z}_j) \mathbb{E}_{q_{-j}} [\log p(\mathbf{z}_j | \mathbf{z}_{-j}, \mathbf{x})] d\mathbf{z}_j - \int q(\mathbf{z}_j) \log q(\mathbf{z}_j) d\mathbf{z}_j$$

- Note here that we have decomposed the expectation over  $q$  as an integral over  $\mathbf{z}_j$  of an expectation over  $q(\mathbf{z}_{-j})$



# Optimizing the ELBO in Mean Field Variational Inference

- To find this argmax, we take the derivative of  $\mathcal{L}_j$  with respect to  $q(z_j)$ , use Lagrange multipliers, and set the derivative to zero:

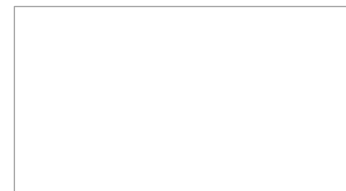
$$\frac{d\mathcal{L}_j}{dq(z_j)} = \mathbb{E}_{q_{-j}} [\log p(z_j | z_{-j}, x)] - \log q(z_j) - 1 = 0$$

- From this, we arrive at the coordinate ascent update:

$$q^*(z_j) \propto \exp \left\{ \mathbb{E}_{q_{-j}} [\log p(z_j | z_{-j}, x)] \right\}$$

- However, since the denominator of the conditional does not depend on  $z_j$ , we can equivalently write:

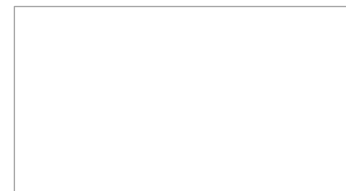
$$q^*(z_j) \propto \exp \left\{ \mathbb{E}_{q_{-j}} [\log p(z_j, z_{-j}, x)] \right\}$$



# Optimizing the ELBO in Mean Field Variational Inference

## Notes:

- This coordinate ascent procedure convergences to a **local maximum**.
- The coordinate ascent update for  $q(\mathbf{z}_j)$  only depends on the other, fixed approximations  $q(\mathbf{z}_k), k \neq j$ .
- While this determines the optimal  $q(\mathbf{z}_j)$ , we haven't yet specified the form (i.e. what specific distribution family) of  $q$  **we aim to use, only the factorization**.
- Depending on what from we use, the coordinate update  $q^*(\mathbf{z}_j)$  might not be easy to work with (and might not be in the same form as  $q(\mathbf{z}_j \dots)$ ).
  - But in many cases it is!
  - And we will specify what forms yield good coordinate updates.



# Optimizing the ELBO in Mean Field Variational Inference

## Simple Example : multinomial conditionals

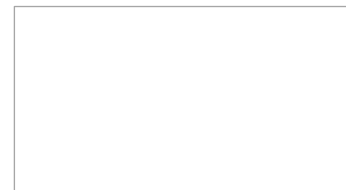
- Suppose we have chosen a model whose conditional distribution is a multinomial, i.e.

$$p(z_j | z_{-j}, x) = \pi(z_{-j}, x)$$

- Then the optimal (coordinate update for)  $q(z_j)$  is :

$$q^*(z_j) \propto \exp\{\mathbb{E}[\log \pi(z_{-j}, x)]\}$$

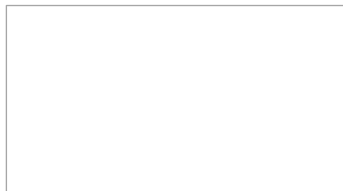
- Which is also a multinomial, and is easy to compute, So choosing a multinomial family of approximations for each latent variable gives closed form coordinate ascent updates.



## Quick Recap

### Quick recap on what we've covered:

- We defined a family of approximations called “mean field” approximations, in which there are no dependencies between latent variables (and also a generalized version of this)
- We decomposed the ELBO into a nice form under mean field assumptions.
- We derived coordinate ascent updates to iteratively optimize each local variational approximation under mean field assumptions.
- Next, we will discuss specific forms for the local variational approximations in which we can easily compute (Closed-form) coordinate ascent updates.



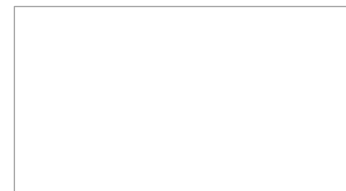
# Exponential Family Conditionals

- Is there a general form for models in which the coordinate updates in mean field variational inference are easy to compute and lead to closed-form updates?
- Yes : the answer is exponential family conditionals.
- i.e., models with conditional densities that are in an exponential family, i.e. of the form :

$$p(z_j | z_{-j}, x) = h(z_j) \exp \left\{ \eta(z_{-j}, x)^T t(z_j) - a(\eta(z_{-j}, x)) \right\}$$

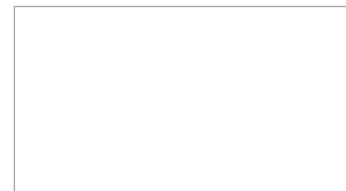
Where  $h, \eta, t$ , and  $a$  are functions that parameterize the exponential family.

- Different choices of these parameters lead to many popular densities (normal, gamma, exponential, Bernoulli, Dirichlet, categorical, beta, Poisson, geometric, etc.).



# Exponential Family Conditionals

- **We call these “exponential-family-conditional” models.**
  - Also known as “conditionally conjugate models”.
- **Many popular models fall into this category, including:**
  - Bayesian mixtures of exponential family models with conjugate priors.
  - Hierarchical hidden Markov models.
  - Kalman filter models and switching Kalman filters.
  - Mixed-membership models of exponential families.
  - Factorial mixtures / hidden Markov models of exponential families.
  - Bayesian linear regression.
  - Any model containing only conjugate pairs and multinomials.
- **Some popular models do not fall into this category, including :**
  - Bayesian logistic regression and other nonconjugate Bayesian generalized linear models:
  - Correlated topic model, dynamic topic model.
  - Discrete choice models.
  - Nonlinear matrix factorization models.





# Exponential Family Conditionals

- We can derive a general formula for the coordinate ascent update for all exponential-family-conditional models.
- First, we will choose the form of our local variational approximation  $q(z_j)$  to be the same as the conditional distribution (i.e. in an exponential family)
- When we perform our coordinate ascent update, we will see that the update yields an optimal  $q(z_j)$  in the same family.
- Recall from above that we derived the coordinate ascent updates for optimizing the ELBO (under the mean field assumption) as :

$$q^*(z_j) \propto \exp\{\mathbb{E}_{q_{-j}}[\log p(z_j | z_{-j}, x)]\}$$

# Exponential Family Conditionals

**Coordinate ascent updated for exponential-family-conditional models (under the mean field approximation):**

- The log of the conditional :

$$\log p(z_j | z_{-j}, x) = \log h(z_j) + \eta(z_{-j}, x)^T t(z_j) - a(\eta(z_{-j}, x))$$

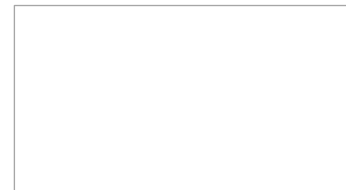
- The expectation of this with respect to  $q(z_{-j})$  is :

$$\mathbb{E}_{q_{-j}}[\log p(z_j | z_{-j}, x)] = \log h(z_j) + \mathbb{E}_{q_{-j}}[\eta(z_{-j}, x)]^T t(z_j) - \mathbb{E}_{q_{-j}}[a(\eta(z_{-j}, x))]$$

- The last term does not depend on  $q(z_j)$ , so we have the update.

$$q^*(z_j) \propto h(z_j) \exp\{\mathbb{E}_{q_{-j}}[\eta(z_{-j}, x)]^T t(z_j)\}$$

- So the optimal  $q(z_j)$  is in the same exponential family as the conditional.



# Exponential Family Conditionals

## Writing this update in terms of variational parameters $v$

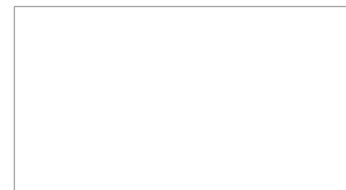
- Give each latent variable a variational parameter  $v_j$ . Under the mean field assumption, we can write the full approximation as :

$$q(z_{1:m}|v) = \prod_{j=1}^m q(z_j | v_j)$$

Where each local variational approximation has an exponential family form.

- Then the coordinate ascent algorithm updates each variational parameter, in turn , as :

$$v_j^* = \mathbb{E}_{q_{-j}}[\eta(z_{-j}, x)]$$



## Quick Recap

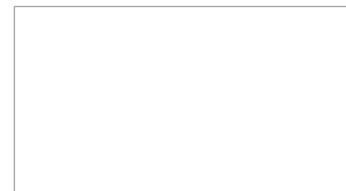
Quick recap on what we've covered :

**We found a family of models (exponential-family-conditional models) in which we have closed form coordinate ascent updates to optimize the ELBO**

- And we gave a number of examples (and non-examples) of these models.

**We gave an explicit for the coordinate ascent update for these exponential-family-conditional models.**

- And also looked at the update in terms of the local variational parameters.



# Mean-Field VI: A Very Simple Example

- Consider data  $\mathbf{X} = \{x_1, \dots, x_N\}$  from a 1-D Gaussian  $\mathcal{N}(x|\mu, \tau^{-1})$  with mean  $\mu$ , precision  $\tau$
- Assume the following normal-gamma prior on  $\mu$  and  $\tau$

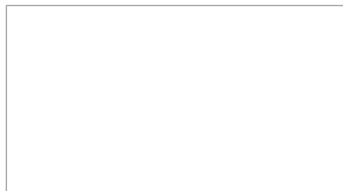
$$p(\mu|\tau) = \mathcal{N}(\mu|\mu_0, (\lambda_0 \tau)^{-1}) \quad p(\tau) = \text{Gamma}(\tau|a_0, b_0)$$

- Note: Here posterior is straightforward (normal-gamma due to the jointly conjugate prior)
- Let's try mean-field VI nevertheless to illustrate the idea
- With mean-field assumption on the variational posterior  $q(\mu, \tau) = q_\mu(\mu)q_\tau(\tau)$

$$\begin{aligned} \log q_\mu^*(\mu) &= \mathbb{E}_{q_\tau} [\log p(\mathbf{X}, \mu, \tau)] + \text{const} \\ \log q_\tau^*(\tau) &= \mathbb{E}_{q_\mu} [\log p(\mathbf{X}, \mu, \tau)] + \text{const} \end{aligned}$$

- In this example, the **log-joint**  $\log p(\mathbf{X}, \mu, \tau) = \log p(\mathbf{X}, \mu, \tau) + \log p(\mu|\tau) + \log p(\tau)$ . Therefore

$$\begin{aligned} \log q_\mu^*(\mu) &= \mathbb{E}_{q_\tau} [\log p(\mathbf{X}, \mu, \tau) + \log p(\tau) + \text{const}] \quad (\text{Only keeping terms that involves } \mu) \\ \log q_\tau^*(\tau) &= \mathbb{E}_{q_\mu} [\log p(\mathbf{X}, \mu, \tau) + \log p(\mu|\tau) + \log p(\tau) + \text{const}] \end{aligned}$$



# Mean-Field VI: A Very Simple Example (Contd)

- Substituting the expressions  $p(\mathbf{X}|\mu, \tau) = \prod_{n=1}^N p(x_n|\mu, \tau)$  and  $\log p(\mu|\tau)$ , we get

$$\begin{aligned}\log q_{\mu}^*(\mu) &= \mathbb{E}_{q\tau}[\log p(\mathbf{X}|\mu, \tau) + \log p(\mu|\tau)] + \text{const} \\ &= -\frac{\mathbb{E}_{q\tau}[\tau]}{2} \left\{ \sum_{n=1}^N (x_n - \mu)^2 + \lambda_0 (\mu - \mu_0)^2 \right\} + \text{const}\end{aligned}$$

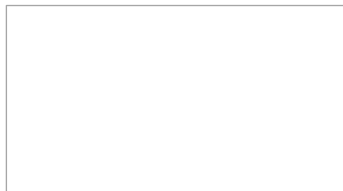
- (Verify) The above is log of a Gaussian. Thus  $q_{\mu}^*(\mu) = \mathcal{N}(\mu|\mu_N, \tau_N)$  with

$$\mu_N = \frac{\lambda_0 \mu_0 + N\bar{x}}{\lambda_0 + N} \quad \text{and} \quad \tau_N = (\lambda_0 + N) \mathbb{E}_{q\tau}[\tau]$$

- Proceeding in a similar way (verify), we can show that  $q_{\tau}^*(\tau) = \text{Gamma}(\tau|a_N, b_N)$

$$a_N = a_0 + \frac{N+1}{2} \quad \text{and} \quad b_N = b_0 + \frac{1}{2} \mathbb{E}_{q\mu}[\sum_{n=1}^N (x_n - \mu)^2 + \lambda_0 (\mu - \mu_0)^2]$$

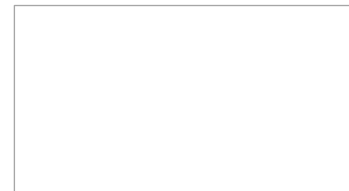
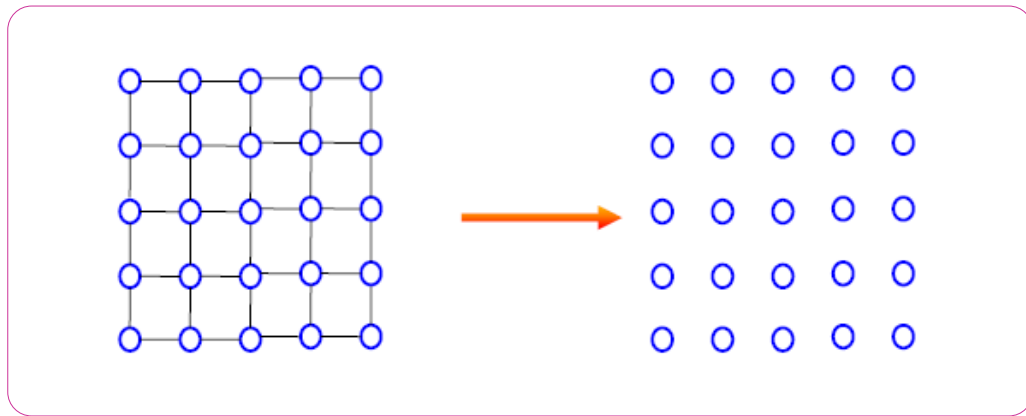
- Important: Updates of  $q_{\mu}^*(\mu)$  and  $q_{\tau}^*(\tau)$  depend on each-other (thus requires cyclic updates)



# Mean Field for Markov Random Fields

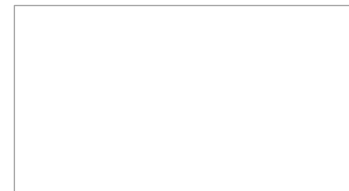
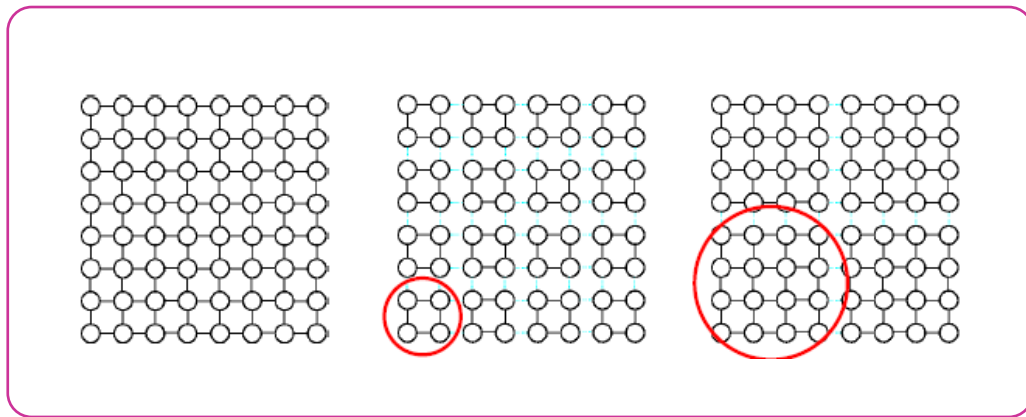
- We can also apply similar mean field approximations for Markov random fields (such as the Ising model):

$$q(x) = \prod_{s \in V} q(x_s)$$



# Mean Field for Markov Random Fields

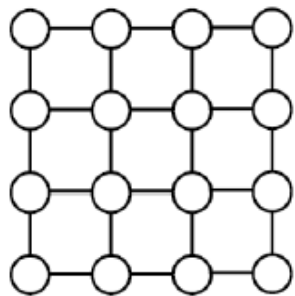
- We can also apply more general forms of mean field approximations (involving clusters) to the Ising models :
- Instead of making all latent variables independent (i.e. naive mean field, previous figure), clusters of (disjoint) latent variables are independent.



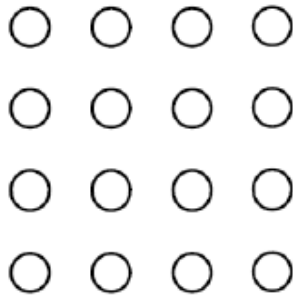


# Structured variational

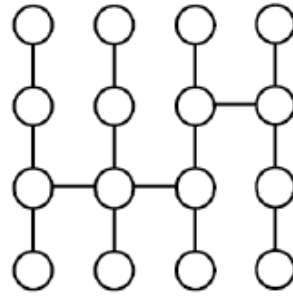
- Mean Field
  - Naive mean field
  - Structured mean field



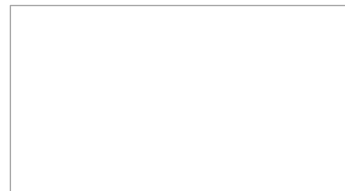
Original Graph



Naïve Mean Field



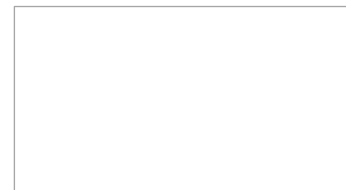
Structured Mean Field



# Local computation of ELBO for factorized P

$$\begin{aligned}
 P(x) &= \frac{1}{Z} \prod_{f_a \in \mathcal{F}} f_a(\mathbf{x}_a) \\
 D(Q||P) &= -H[Q] - E_Q \left[ \log \left( \frac{1}{Z} \prod_{f_a \in \mathcal{F}} f_a(\mathbf{x}_a) \right) \right] \\
 &= -H[Q] - \log \frac{1}{Z} - \sum_{f_a \in \mathcal{F}} E_Q [\log f_a(\mathbf{x}_a)] \\
 &= \log Z - \left( H[Q] + \sum_{f_a \in \mathcal{F}} E_Q [\log f_a(\mathbf{x}_a)] \right)
 \end{aligned}$$

$$\mathcal{L}[Q] = H[Q] + \sum_{f_a \in \mathcal{F}} E_Q [\log f_a(\mathbf{x}_a)]$$



# Naïve mean field for factorized P

Naïve mean field (i.e., fully factored distribution  $Q$ ) :

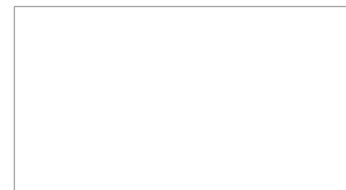
$$Q(x) = \prod_{i=1}^N Q_i(x_i)$$

$$\mathcal{L}[Q] = \sum_{a \in \mathcal{F}} E_Q[\log f_a(\mathbf{x}_a)] + H[Q]$$

$$E_Q[\log f_a(\mathbf{x}_a)] = \sum_{x_a \in \text{Val}(X_a)} \left( \prod_{i \in \mathcal{N}(a)} Q_i(x_i) \right) \log f_a(\mathbf{x}_a)$$

$$H[Q] = \sum_{i=1}^N H[Q_i] \quad \mathcal{N}(a) = \{i | x_i \in \text{scope}(f_a)\}$$

- Thus,  $\mathcal{L}[Q]$  can be rewritten simply as a sum of expectations, each one over a small set of variables



# Stationary point (fixed-point equations)

$$Q_i(x_i) = \frac{1}{z_i} \exp \left\{ \sum_{a: i \in \mathcal{N}(a)} \sum_{x_a \in \text{Val}(x_a)} Q(x_a | x_i) \log f_a(x_a) \right\}$$

- Proof:

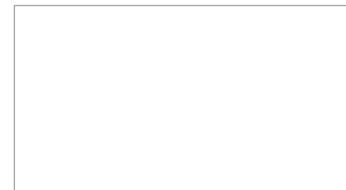
$$\mathcal{L}[Q] = \sum_{i=1}^N \mathcal{L}_i[Q]$$

$$\mathcal{L}_i[Q] = \sum_{a: i \in \mathcal{N}(a)} \sum_{x_a} \left( \prod_{j \in \mathcal{N}(a)} Q_j(x_j) - 1 \right) \log f_a(x_a) + H[Q_i]$$

$$L_i[Q, \lambda] = \mathcal{L}_i[Q] + \lambda_i \left( \sum_{x_i \in \text{Val}(x_i)} Q_i(x_i) - 1 \right)$$

$$\frac{\partial L_i}{\partial Q_i(x_i)} = 0 \Rightarrow Q_i(x_i) = \frac{1}{e^{1-\lambda_i}} \exp \left\{ \sum_{a: X_i \in \mathcal{N}(a)} \sum_{x_a} Q(x_a | x_a) \log f_a(x_a) \right\}$$

Update rule: We can optimize each  $Q_i$  given values for other potentials



# Optimization by coordinate ascent for factorized P

$$Q_i(x_i) = \frac{1}{z_i} \exp \left\{ \sum_{a: X_i \in \mathcal{N}(a)} \sum_{x_a \in \text{Val}(X_a)} Q(x_a | x_i) \log f_a(x_a) \right\}$$

All these terms involve expectations of variables other than  $X_i$  and do not depend on the choice of  $Q_i(X_i)$ .

- Coordinate ascent algorithm repeatedly optimizes a single marginal at a time, given fixed choices to all of the others.

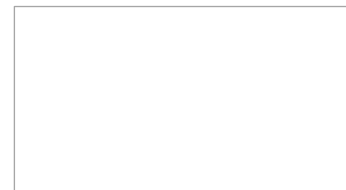
While not converged

Iterate over each of the variables  $i \in \mathcal{V}$

Maximize the objective function with respect to

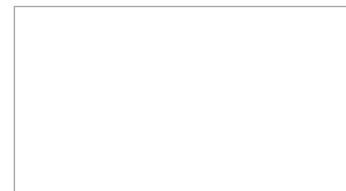
$Q_i(x_i), \forall x_i \in \text{Val}(X_i)$  by the above formula.

Block coordinate descent



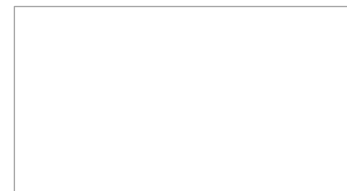
# Convergence properties

- $\mathcal{L}_i$  is concave in  $Q_i(X_i)$ 
  - Update of  $Q_i$  is guaranteed to increase (or not decrease)  $\mathcal{L}$
- Mean Field iterations are guaranteed to converge.
  - Each step of coordinate ascent procedure is monotonically non-decreasing in  $\mathcal{L}$ .
  - Because  $\mathcal{L}$  is bounded, the sequence of distributions represented by successive iterations of Mean-Field must converge.
- At the convergence point, the fixed-point equations hold for all variables.
  - As a consequence, the convergence point is a stationary point of the energy functional subject to the constraints.
- The result of the mean field approximation is a local maximum, and not necessarily a global one



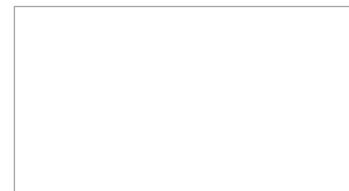
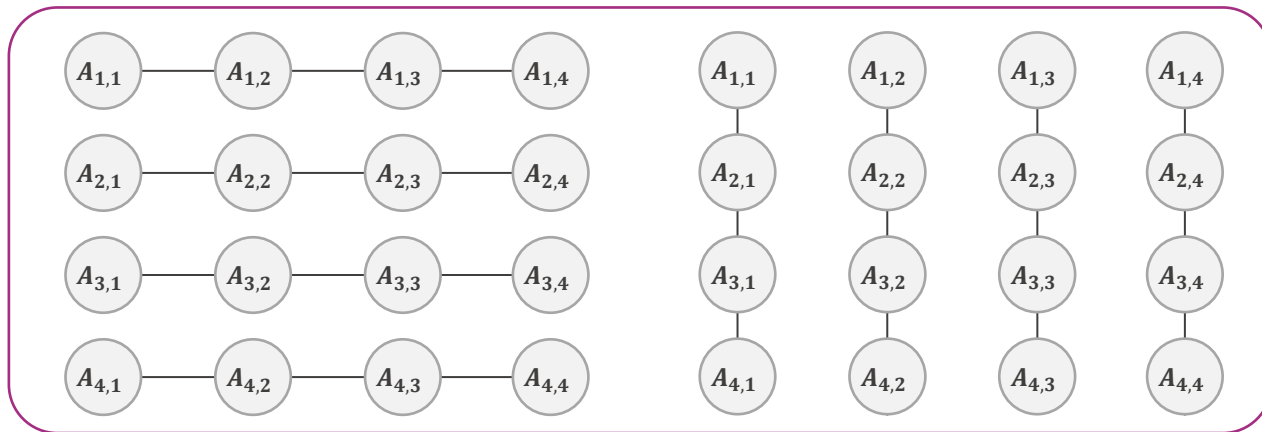
## Local computation in naive mean field

- When updating  $Q_j$ , we only need to reason about the variables which share a factor with  $Z_j$ 
  - The expectations required to evaluate  $Q_j$  involve only those variables lying in the **Markov blanket of the node  $j$** 
    - The other terms get absorbed into the constant term.
- The optimization of  $Q_j$  can therefore be expressed as a local computation at the node



# Structured Mean Field

- Naive mean-field can lead to very poor approximations
  - We must use a richer class of distributions  $Q$ , which has greater expressive power (by capturing some of the dependencies in  $P$ )
    - Use network structures of different complexity
    - Subgraph of  $G_P$  over which exact computation of  $H[Q]$  is feasible
- Example: for grid  $G_P$ , a collection of independent chain structures.
  - Exact inference with such structures is linear





## Example: Mean field for pairwise MRFs

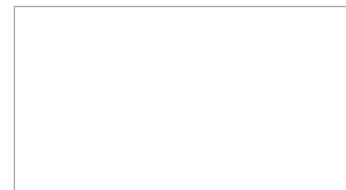
$$P(x) = \frac{1}{Z} \exp \left\{ \sum_{(i,j) \in \mathcal{E}} \theta_{ij}(x_i, x_j) + \sum_{i \in \mathcal{V}} \theta_i(x_i) \right\}$$

$$Q^* = \operatorname{argmax}_{Q \in \mathcal{Q}} \mathcal{L}[Q]$$

Subject to:

$$Q(x) = \prod_{i=1}^N Q_i(x_i)$$

$$\sum_{x_i \in \text{Val}(x_i)} Q_i(x_i) = 1$$



# Example: Mean field for pairwise MRFs

- P: Pairwise MRF

$$P(x) = \frac{1}{Z} \prod_{(i,j) \in \mathcal{E}} \phi_{ij}(x_i, x_j) \prod_{i \in \mathcal{V}} \phi_i(x_i)$$

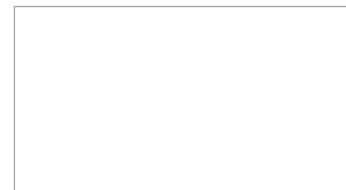
$$P(x) = \frac{1}{Z} \exp \left\{ \sum_{(i,j) \in \mathcal{E}} \theta_{ij}(x_i, x_j) + \sum_{i \in \mathcal{V}} \theta_i(x_i) \right\}$$

$$\begin{aligned} \theta_i &= \ln \phi_i \\ \theta_{ij} &= \ln \phi_{ij} \end{aligned}$$

$$Q_i(x_i) = \frac{1}{Z_i} \exp \left\{ \theta_i(x_i) + \sum_{j \in \mathcal{N}(i)} \sum_{x_j} Q_j(x_j) \theta_{ij}(x_i, x_j) \right\}$$

$$\Rightarrow Q_i(x_i) \propto \phi_i(x_i) \prod_{j \in \mathcal{N}(i)} m_{ji}(x_i)$$

$$m_{ij}(x_i) \propto \exp \left\{ \sum_{x_j} Q_j(x_j) \theta_{ij}(x_i, x_j) \right\}$$



# Structured stationary point

$$P(\mathbf{x}) = \frac{1}{Z} \prod_{k=1}^K \phi_k(\mathbf{x}_k)$$

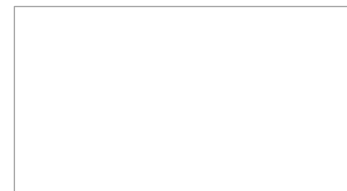
$$Q(\mathbf{x}) = \frac{1}{Z_Q} \prod_{j=1}^J \psi_j(\mathbf{x}_j)$$

$$F[P, Q] = \sum_{k=1}^K E_Q[\ln \phi_k(\mathbf{x}_k)] - E_Q[\ln Q]$$

$$F[P, Q] = \sum_{k=1}^K E_Q[\ln \phi_k(\mathbf{x}_k)] - \sum_{j=1}^J E_Q[\ln \psi_j(\mathbf{x}_j)] + \ln Z_Q$$

## Structured mean-field quality

- Both the quality and the computational complexity of the variational approximation depend on the structure of  $P$  and  $Q$ .
- We want to be able to perform efficient inference in the approximating network.
- We often select our network so that the resulting factorization leads to a tractable network (that is, one of low tree-width)



# Structured stationary point

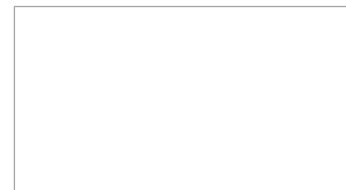
- $\psi_j$  is a stationary point of the energy functional iff:

$$\psi_j(x_j) \propto \exp \left\{ E_Q \left[ \log \tilde{P}(x) | x_j - \sum_{k \neq j} E_Q [\log \psi_k(x_k) | x_j] \right] \right\}$$

$$\psi_j(x_j) \propto \exp \left\{ \sum_i E_Q \left[ \log \phi_i(x_i) | x_j - \sum_{k \neq j} E_Q [\log \psi_k(x_k) | x_j] \right] \right\}$$

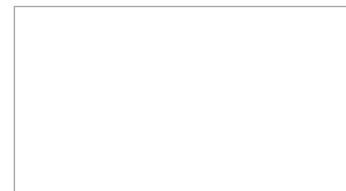
$\psi_j(x_j)$  dose not affect the right-hand side of the fixed-point equations defining its value

- We need to perform inference after each update step

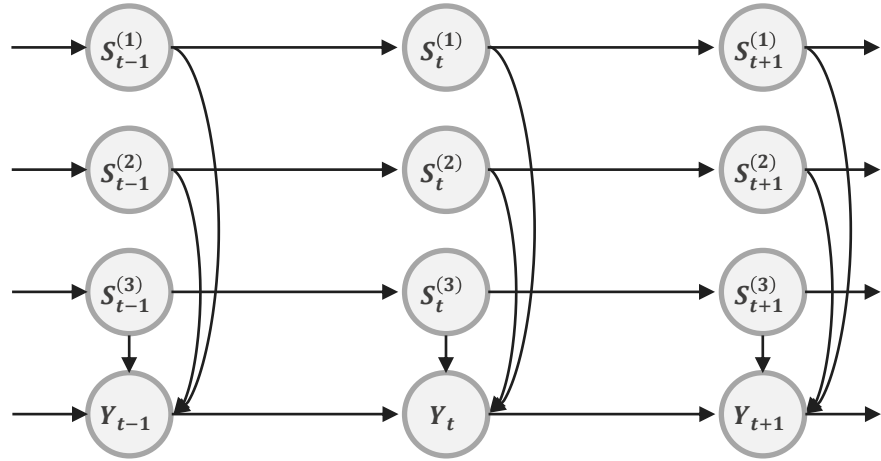
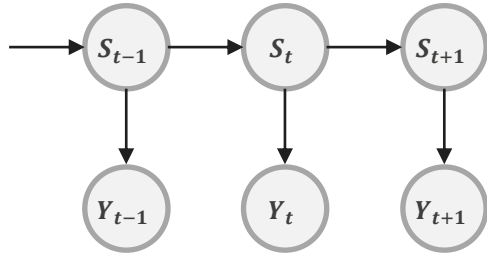


# Generalized (Cluster-based) Mean Field for MRFs

- For these MRFs there exist a general, iterative message passing algorithm for inference (similar to the loopy-BP algorithm learned in the previous class).
- Clustering completely defines the approximation.
  - Preserves Dependencies
  - Allows for a flexible performance/cost trade-off.
  - Clustering can be done in an automated fashion.
- Generalizes model-specific structured VI algorithms, including :
  - fHMM, LDA.
  - Variational Bayesian learning algorithms
- Provides new structure VI approximations to complex models



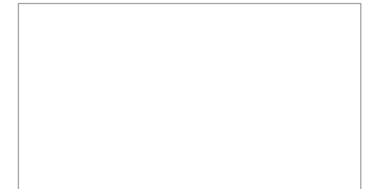
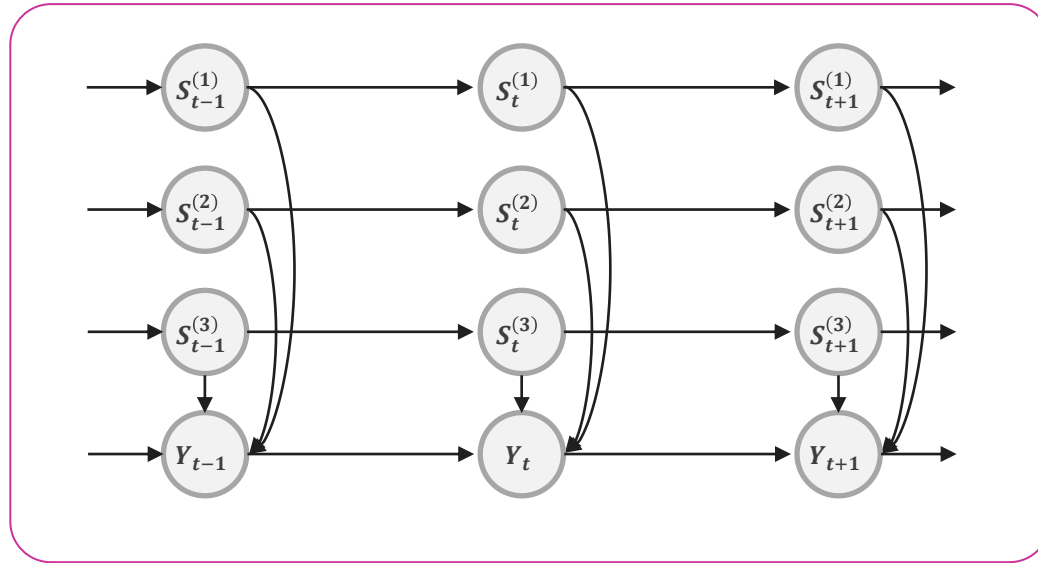
# Example: Factorial HMM



$$P(S_t | S_{t-1}) = \prod_{m=1}^M P(S_t^{(m)} | S_{t-1}^{(m)})$$

# Factorial HMM: Exact Inference

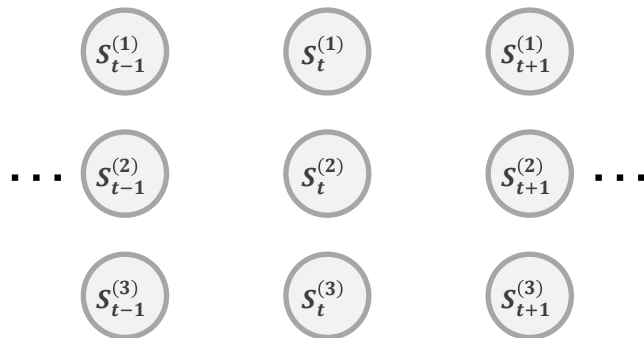
- $O(TMK^{M+1})$  for inference of a single observation sequence
- $T$ -Sequence length,  $M$ =#of states,  $K$ =Cardinality of each state
- We can use variational inference for efficient parameter estimation (EM) as well as inference





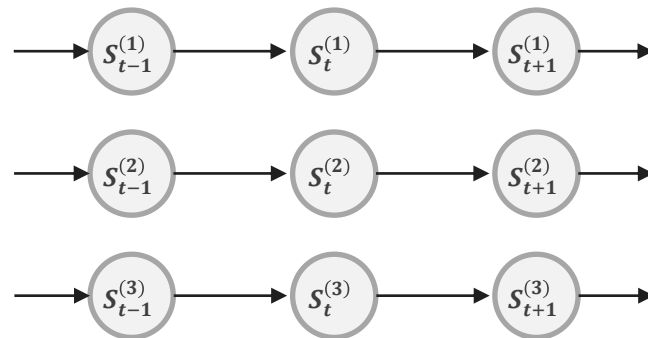
# Factorial HMM: Structured Variational Inference

(a)



$$Q(\{S_t\}|\theta) = \prod_{t=1}^T \prod_{m=1}^M Q(s_t^{(m)}|\theta_t^{(m)})$$

(b)

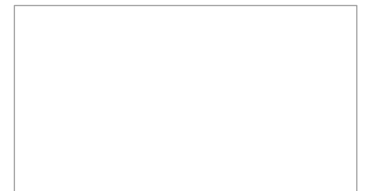


$$Q(\{S_t\}|\theta) = \frac{1}{Z_Q} \prod_{m=1}^M Q(s_1^{(m)}|\theta) \prod_{t=2}^T Q(s_t^{(m)}|s_{t-1}^{(m)}, \theta)$$

- $O(TMK^2)$  per iteration of EM for updating latent states in (a)
- Can you figure out the cost per iteration in (b) ?

# Bayesian Inference

- Bayesian Modelling
- Priors
- MLE and MAP
- Bayesian Inference Examples
- Tractability and Approximation Methods
- Variational Inference
- Evidence Lower Bound (ELBO)
- Kullback-Liebler (KL) Divergence
- Mean field Variational Inference
- **Variational Bayes Learning**
- Myths about Bayesian Methods



# Lower Bounding the Marginal Likelihood Variational Bayesian Learning

Let the latent variables be  $z$ , observed data  $x$  and the parameters  $\theta$ .

We can **lower bound** the **marginal likelihood** (Jensen's inequality) :

$$\begin{aligned}\ln p(x|m) &= \ln \int p(x, z, \theta|m) dz d\theta \\ &= \ln \int q(z, \theta) \frac{p(x, z, \theta|m)}{q(z, \theta)} dz d\theta. \\ &\geq \int q(z, \theta) \ln \frac{p(x, z, \theta|m)}{q(z, \theta)} dz d\theta.\end{aligned}$$

Use a simpler, factorized approximation for  $q(z, \theta) \approx q_z(z)q_\theta(\theta)$ :

$$\begin{aligned}\ln p(x|m) &\geq \int q_z(z)q_\theta(\theta) \ln \frac{p(x, z, \theta|m)}{q_z(z)q_\theta(\theta)} dz d\theta \\ &\stackrel{\text{def}}{=} \mathcal{F}_m(q_z(z), q_\theta(\theta), x).\end{aligned}$$

# Variational Bayesian Learning...

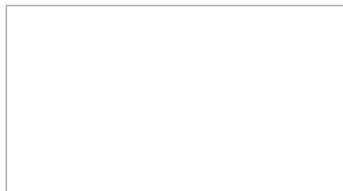
Maximizing this **lower bound**,  $\mathcal{F}_m$ , leads to **EM-like** iterative updates:

$$\begin{aligned} q_z^{(t+1)}(z) &\propto \exp \left[ \int \ln p(z, x | \theta, m) q_\theta^{(t)}(\theta) d\theta \right] && \text{E-like step} \\ q_\theta^{(t+1)}(\theta) &\propto p(\theta | m) \exp \left[ \int \ln p(z, x | \theta, m) q_z^{(t+1)}(z) dz \right] && \text{M-like step} \end{aligned}$$

Maximizing  $\mathcal{F}_m$ , is equivalent to minimizing KL-divergence between the **approximate posterior**,  $q_\theta(\theta) q_z(z)$  and the **exact posterior**,  $p(\theta, z | x, m)$  :

$$\ln p(x | m) - \mathcal{F}_m(q_z(z), q_\theta(\theta), x) = \int q_z(z) q_\theta(\theta) \ln \frac{q_z(z) q_\theta(\theta)}{p(\theta, z | x, m)} dz d\theta = \mathbf{KL}(q \parallel p)$$

In the limit as  $n \rightarrow \infty$ , for identifiable models, the variational lower bound approaches the BIC criterion.



# The Variational Bayesian EM algorithm

## EM for MAP estimation

**Goal:** maximize  $p(\theta|x, m)$  w.r.t.  $\theta$

**E Step:** compute

$$q_z^{(t+1)}(z) = p(z|x, \theta^{(t)})$$

**M Step:**

$$\theta^{(t+1)} = \operatorname{argmax}_{\theta} \int q_z^{(t+1)}(z) \ln p(z, x, \theta) dz$$

## Variational Bayesian EM

**Goal:** Lower bound  $p(x|m)$

**VB-E Step:** compute

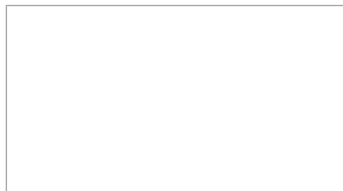
$$q_z^{(t+1)}(z) = p(z|x, \bar{\phi}^{(t)})$$

**VB-M Step:**

$$q_{\theta}^{(t+1)}(\theta) \propto \exp \left[ \int q_z^{(t+1)}(z) \ln p(z, x, \theta) dz \right]$$

### Properties:

- Reduces to the EM algorithm if  $q_{\theta}(\theta) = \delta(\theta - \theta^*)$ .
- $\mathcal{F}_m$  increases monotonically, and incorporates the model complexity penalty.
- Analytical parameter distributions (But not constrained to be Gaussian).
- VB-E step has same complexity as corresponding E step.
- We can use the junction tree, belief propagation, Kalman filter, etc., algorithms in the VB-E step of VB-EM, but **using expected natural parameters,  $\bar{\phi}$**



# Variational Bayesian EM

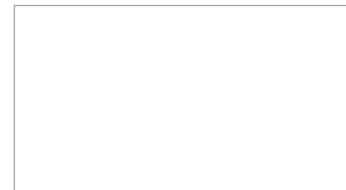
The Variational Bayesian EM algorithm has been used to approximate Bayesian learning in a wide range of models such as:

- Probabilistic PCA and factor analysis
- Mixtures of Gaussians and mixtures of factor analysers
- Hidden Markov models
- State-space models (Linear dynamical systems)
- Independent components analysis (ICA)
- Discrete graphical models...

The main advantage is that it can be used to **automatically do model selection** and does not suffer from overfitting to the same extent as ML methods do.

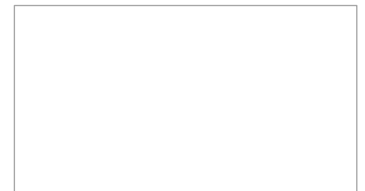
Also it is about as computationally demanding as the usual EM algorithm.

See: [www.variational-bayes.org](http://www.variational-bayes.org)



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# Myths and misconceptions about Bayesian methods

## Bayesian method make assumptions where other methods don't

- All methods make assumptions! Otherwise it's impossible to predict. Bayesian methods are transparent in their assumptions whereas other methods are often opaque.

## If you don't have the right prior you won't do well

- Certainly a poor model will predict poorly but there is no such thing as the right prior! Your model (both prior and likelihood) should capture a reasonable range of possibilities. When in doubt you can choose vague priors (cf nonparametric)

## Bayesian methods are generative

- You can use Bayesian approaches for both generative and discriminative learning (e.g. Gaussian process classification, Conditional Random Fields).

## Bayesian methods don't scale well

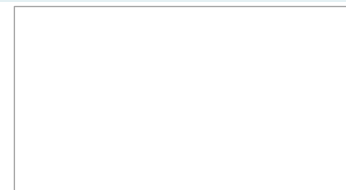
- With the right inference methods (variational, MCMC) it is possible to scale to very large datasets (e.g. excellent results for Bayesian Probabilistic Matrix Factorization on the Netflix data using MCMC), but it's true that averaging/integration is often more expensive than optimization.

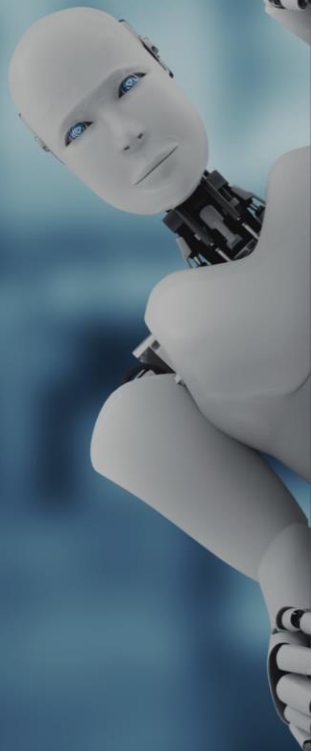




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**Trigonometry**

$\sin \theta = \frac{\text{OPP}}{\text{HYP}}$   
 $\cos \theta = \frac{\text{ADJ}}{\text{HYP}}$   
 $\tan \theta = \frac{\text{OPP}}{\text{ADJ}}$

$\sin^2 \theta + \cos^2 \theta = 1$   
 $\sec^2 \theta = 1 + \tan^2 \theta$   
 $\csc^2 \theta = 1 + \cot^2 \theta$

$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}$   
 $\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$   
 $\tan \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$

**Calculus**

$\oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt}$   
 $\oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$   
 $\oint \vec{E} \cdot d\vec{l} = (E + dE)h_f - E h_f = h_f dE$   
 $\phi_B = BA = Bh_f dx$   
 $\frac{d\phi_B}{dt} = h_f dx \frac{dB}{dt}$   
 $\oint \vec{B} \cdot d\vec{l} = -h_f dB$   
 $\oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_E}{dt}$   
 $h_f dE = -h_f dx \frac{dB}{dt}$   
 $\frac{dE}{dx} = -\frac{dB}{dt}$

**Electronics**

**PNP Transistor Circuit**

$I_E = I_B + I_C$  and  $\frac{I_C}{I_E} = \alpha$   
 $I_B = I_E - I_C$   
 $I_B = I_E (1 - \alpha)$   
 $\beta = \frac{I_C}{I_B} = \frac{\alpha}{1 - \alpha}$

**AC Source**

$P_o = V_o I_o \cos \theta_o$   
 $\cos \theta_o = \frac{P_o}{V_o I_o}$   
 $I_c = I_o \cos \theta_o$ ,  $I_m = I_o \sin \theta_o$   
 $R_c = \frac{V_o^2}{P_o}$ ,  $X_m = \frac{V_o}{I_m}$

**Optics**

$m = \frac{f}{s - f} = \frac{s' - f}{f}$   
 $\frac{1}{f} = \frac{1}{s} + \frac{1}{s'}$   
 $m = \frac{i}{o} = \frac{s'}{s}$   
 $f = \frac{R}{2}$

**Chemistry**

**Photosynthesis**

$6CO_2 + 12H_2O \xrightarrow{\text{light Chlorophyll}} C_6H_{12}O_6 + 6O_2$

**Protein Synthesis**

**1-6 DIHEXANOIC ACID**

$n \text{ HOOC(CH}_2)_4\text{COOH} + n \text{ H}_2\text{N(CH}_2)_4\text{NH}_2 \rightarrow \text{NYLON 6-6}$

**2AIBr<sub>3</sub> + 3Cl<sub>2</sub> → 2AI**

**ETHANOL**

**ETHYL PROPANOATE**

**Mathematics**

$\rho = \frac{m}{V}$   
 $\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$   
 $y = A \sin(\omega t - kx)$   
 $k = \frac{2\pi}{\lambda}$ ,  $\frac{1}{v} = \frac{\mu_0 \epsilon_0}{\omega}$   
 $E = E_{\max} \sin(\omega t - kx)$

**Set Theory**

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$   
 $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$

Thank you Q&A